

Honeywell

Experion
Backup and Restore
User's Guide

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About This Document

This guide describes how to use Experion Backup and Restore (EBR) in a Honeywell process control system. You use EBR to back up and then restore critical system data for TotalPlant Solution (TPS) system and Experion Process Knowledge System (PKS) nodes after recovery from a node failure.

Intended audience

This guide is intended for engineers, system administrators, and other technical personnel who are responsible for backing up and restoring data for TPS and Experion PKS nodes.

Prerequisite skills

You must have practical experience with system backup and restore operations, as well as proficiency with the following Microsoft® Windows operating systems:

- Windows Server 2003™ Standard
- Windows XP™ Professional (SP2 or later)

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References

The following list identifies all documents that may be sources of reference for material mentioned in this publication.

Document	Description
<i>Server and Client Planning Guide</i>	Contains high-level planning and design topics for Experion PKS servers and clients, and for controllers other than Process Controllers.
<i>Server and Client Configuration Guide</i>	Contains information about filtering system events.
<i>Control Building Guide</i>	Contains control strategy checkpoints.
• <i>Symantec Backup Exec™ System Recovery Manager Implementation Guide</i> • <i>Symantec Backup Exec™ System Recovery</i>	Contains information about installing, using, and troubleshooting EBR. Both guides are included on the EBR product DVD.

Abbreviations

The following list defines the special abbreviations used in this guide.

Abbreviation	Definition
ACE	Application Control Environment
APP	Advanced Processing Platform
EBR	Experion Backup and Restore
EMDB	Enterprise Model Database
ERDB	Engineering Repository Database
FTE	Fault-Tolerant Ethernet
GUS	Global User Station
MBR	Master Boot Record
PHD	Process History Database
QVCS	Qualification and Version Control System
SIM	Simulation Interface Module
TPN	TotalPlant Solution (TPS) Network
TPS	TotalPlant Solution System

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Training Classes	Honeywell Automation College: http://www.automationcollege.com		

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE - EXTERNAL: Identifies an additional source of information outside of the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	STOP: Indicates that your task is complete. Return to the checklist or procedure that led to this task.

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Symbol Definitions

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1. Introduction to EBR

1.1 EBR functions

Experion Backup and Restore (EBR) provides a real-time, continuous backup infrastructure for TPS and Experion PKS nodes within a Honeywell process control system.

During a backup (which might be scheduled or manual), EBR copies the disk image of each “Target” node (a node with EBR installed) to a Backup Repository (usually a file server). During a restore, the disk image restores the hard disk data to the Target Nodes and Experion repositories [Enterprise Model Database (EMDB) and Engineering Repository Database (ERDB)].

EBR provides the following functionality:

- **Graphical user interfaces (GUI)** are used for creating backup jobs and monitoring job status through various event logs and status notifications. Depending on the type of EBR installation, three distinct GUIs are available—EBR Manager Console for centrally managed environments, and EBR Server User Interface and EBR Desktop User Interface for locally managed environments.
- The **rapid restore** function uses a bootable CD/DVD that contains all the hard disk and network drivers required to quickly restore a repaired or replacement node.
- **Disaster recovery backup** provides selections for setting scheduled or manual backup jobs. The resulting backup is used restoring the original node or replacement node (to when it was last backed up). No post-restore installation or configuration is required. Disaster recovery backup jobs in addition perform a sector-by-sector backup of all node data including the operating system, drivers, service packs, Experion and/or TPS data, and node configuration.
- **File/folder recovery** provides selections for a full or incremental backup which must be configured as a read-only drive, thus allowing you to view, copy, compare, or run a virus scan of the backup image.
- **EBR user accounts** must be set up to allow authorized personnel to perform backup jobs without having to grant them administrative permissions.

1.2 EBR terms

The following list defines the EBR terms used in this guide.

- **Recovery point** and **backup image** are synonymous terms for an exact reproduction of the contents of a hard disk. Typically, the backup image contains the applications, configuration settings, operating system, and other data that is compressed and stored as a file. The backup image is used for restoring a hard disk to the state when it was last backed up.
- The **agent** is the EBR software that is installed on a node to create backup images and send them to the backup repository.
- The **Manager Node** is a server computer that uses EBR to manage backup jobs and deploy backup agents to the Target Nodes in a centrally-managed environment. In addition, the Manager Node contains the system configuration information and maintains a database of backup history.
- A **Server Node** is a computer with the EBR agent installed to perform backups, and that uses EBR to manage its backup jobs and deploy backup agents to Target Nodes in a locally-managed environment where EBR Manager is not used. A “Server” is not defined as such by its *function* within the system, but by its *operating system*, which must be Windows Server 2003 Standard.
- A **Desktop** or **Workstation Node** is a computer with the EBR agent installed to perform backup of its hard drive, and that uses EBR to manage its backup jobs in a locally-managed environment. A “Desktop” is not defined as such by its *function* within the system, but by its *operating system*, which must be Windows XP Professional.
- A **Target Node** or **Client** is a computer with EBR installed (must be a Server or Desktop Node).
- The **Backup Repository** or **backup destination** is a computer (usually a file server) that stores all of the backup images for the Target Nodes.
- EBR provides three distinct GUIs to create and manage backup jobs:
 - **EBR Manager Console**: Typically installed on the Manager Node.
 - **EBR Server User Interface**: Installed on a Server Node.
 - **EBR Desktop User Interface**: Installed on a Desktop Node.
- A **Distribution Point** is a computer that holds EBR software modules for installation over the network.

2. Installing and configuring EBR

2.1 About this section

This section describes how to install and configure EBR to back up and restore your TPS and/or Experion Nodes in a Honeywell Process Control system.

2.2 Preparing to install EBR

Previous versions of EBR or Symantec Backup Exec Recovery

If an earlier version of EBR and/or Symantec Backup Exec System Recovery software is installed on the node, you must uninstall them before attempting to install the latest EBR software. For instructions about removing EBR, see Chapter 3, “Removing EBR.”

EBR software packages

EBR contains Server and Desktop software packages designed for installation on nodes with Windows operating systems as specified in the following table:

Software package	Functional description	EBR package components	Specified Windows operating system
Server	Manages backups on the local node and deploys EBR agents to the Target Nodes	Server	Windows Server 2003 Standard
Desktop	Manages backups on the local node	Desktop	Windows XP Professional (SP2 or later)
Manager	Manages the recovery activities and monitors real-time status and quickly resolves any problems identified	Manager	Windows Server 2003 Standard

EBR installation environments

EBR must be installed in a centrally or locally-managed environment.

- In a **centrally-managed environment**, EBR is installed on the Manager Node and the Manager is used for deploying EBR agents to the Target Nodes.
- In a **locally-managed environment**, EBR is installed on each Server or Desktop Node to be backed up.

Minimum platform requirements for a Target Node

EBR supports the following minimum platforms for a Target Node:

- Intel® Pentium processor, 1 GHz (greater than 1 GHz is recommended)
- Microsoft Windows Server 2003 Standard, Windows XP Professional (SP2 or later) operating system
- 1 GB RAM
- Experion PKS R301 and up; TPS R400 and up

FTE considerations

If you use Fault-Tolerant Ethernet (FTE), **Honeywell Fault Tolerant Ethernet Virtual Adapter** errors are reported for EBR Server or Desktop software installations. (These errors do not interfere with a recovery and are expected.) For more information about FTE errors, see Chapter [2.12](#), “Validating Drivers.”

2.3 Checklists for installing EBR



ATTENTION

Use the appropriate checklist in this section to install the applicable EBR software for your particular system. The checklists specify the correct sequence of tasks for a successful installation.



ATTENTION

- The computer used as the Backup Repository must *not* be an Experion server, Process History Database (PHD) server, or Advanced Processing Platform (APP) node.
- You must install the EBR Manager on a dedicated server machine and not on a domain controller.
 - Before installing EBR software on the Manager Node or Server Node, install Microsoft .NET Framework version 2.0 on the node available with the package.

CAUTION

Honeywell recommends that you install EBR on *both* servers of a redundant pair to be backed up. However, if you decide to back up only one of the servers, you must install EBR on Server B because it contains the primary ERDB and EMDB.

Installing EBR over a Network

Step	Task	Go to	Done?
1	Create a Distribution Point.	page 24	
2	Install EBR software packages over a network.	Page 27	
3	Publish EBR software using Group Policy Object.	Page 29	
4	Install EBR on Clients using published software.	Page 31	

Installing EBR on the Manager Node

Step	Task	Go to	Done?
1	Ensure that Microsoft .NET Framework version 2.0 is installed on the Manager Node.	n/a	
2	Install the EBR Manager software package on the Manager Node.	page 32	
Configuring the EBR Manager			
3	Start the Manager.	page 54	
4	Set the backup defaults.	page 54	
5	Copy command files to PHD nodes.	page 56	
6	Copy command files to Experion Servers.	page 56	
7	Configure the Windows Firewall for nodes <i>with</i> Experion software installed.	page 61	
8	Configure the Windows Firewall for nodes <i>without</i> Experion software installed.	page 62	
9	Add node groups (if necessary).	page 57	
10	Add nodes.	page 57	

Installing EBR on a Server Node

Step	Task	Go to	Done?
1	Install the EBR Server software package on each server to be backed up.	page 43	
Configuring the Server User Interface			
2	Start the EBR Server User Interface.	page 59	
3	Set the backup defaults.	page 59	
4	Configure the Windows Firewall for nodes <i>with</i> Experion software installed.	page 61	
5	Configure the Windows Firewall for nodes <i>without</i> Experion software installed.	page 62	

Installing EBR on a Desktop Node

Step	Task	Go to	Done?
1	Install the EBR Desktop software package on each desktop node to be backed up.	page 48	
Configuring the Desktop User Interface			
2	Start the Desktop User Interface.	page 59	
3	Set the backup defaults.	page 59	
4	Configure the Windows Firewall for nodes <i>with</i> Experion software installed.	page 61	
5	Configure the Windows Firewall for nodes <i>without</i> Experion software installed.	Page 62	

2.4 Installing EBR over a network

To begin with, you need to create a Distribution Point in a server computer and a folder structure for the EBR software packages to be installed. The Distribution Point must be accessible on the network as it acts as a host for the EBR packages/components to be installed.

To create Distribution Point

You must create an individual directory for each package that contains an install.exe and setup.exe files. Follow these steps to create a Distribution Point in the server machine.

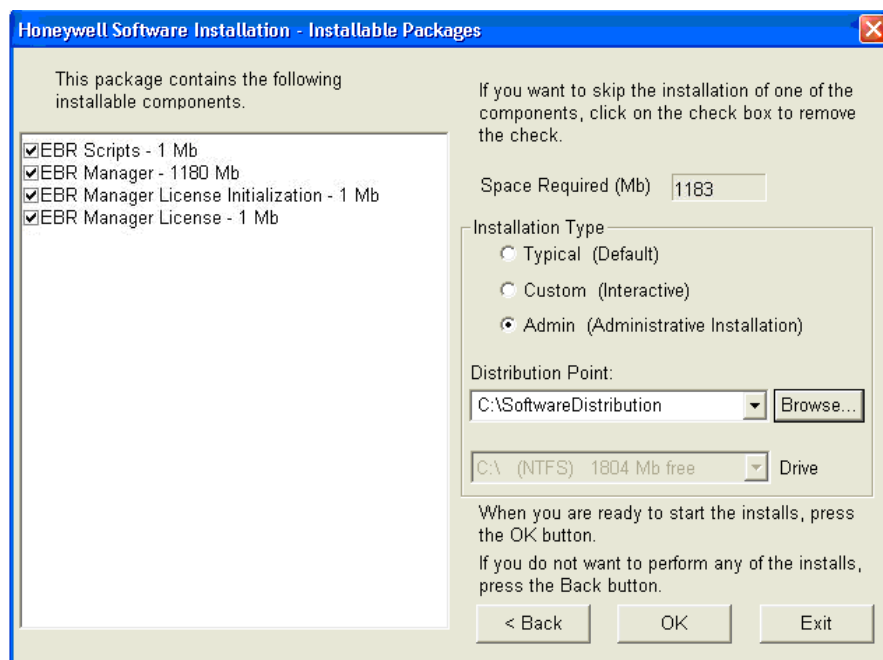
Step	Action
1	Log on to a server machine as a Domain Administrator . For example, the machine name is uis-msi-svr.
2	Click Start and then click My Computer .
3	Double-click drive C .
4	Click File>New>Folder , to create a new folder. Name it as Software Distribution .
5	Open the Software Distribution folder by double-clicking it.
6	Create three new folders as EBR_Manager , EBR_Server and EBR_Desktop .
7	Right -click Software Distribution folder, and then click Properties .
8	Click the Sharing and Security... tab. The Software Distribution Properties page appears.
9	Click Share this folder option.
10	Click Permissions .
11	Click Add... , and then select Administrator .
12	Click OK .
13	Select Administrator , and then click Full Control .
14	On the Permissions for Software Distribution page, highlight Everyone , and then click Remove .

Step	Action
15	Click Add , and then select Authenticated Users .
16	Click Add , and then click OK .
17	Select Authenticated Users , and then select the corresponding check box displayed against Read under Allow . Note: Ensure the permission for Full Control and Change are unchecked.
18	Click Apply , and then click OK .
19	Select Security tab on the Software Distribution Properties page.
20	Click Advanced . The Advanced Security for Software Distribution page appears.
21	Select Everyone , and then click Remove .
22	Clear Allow inheritable permissions from parent to propagate to this object .
23	Click Remove on the Security page.
24	Click Add on Software Distribution Properties page, and then select Authenticated Users .
25	Click Add , and then click OK .
26	Select Authenticated Users , and then select the corresponding check box displayed against Read & Execute, List Folder Contents, and Read under Allow . Note: Ensure the permission for <i>Full Control, Modify</i> and <i>Write</i> are cleared.
27	Click Apply , and then click OK .
	Stop The task is complete. Return to the checklist that led to this task.

Administrative Install

This feature is used for preparing a package to be installed from the network accessible Distribution Point.

Click **Admin (Administrative Installation)**, and then enter a Distribution Point or browse, as shown in the following example:



The Admin Install interaction is automatically recorded in an install.ini file. This file is located on the Distribution Point where the install.exe is located. The install.ini is used for subsequent client installation, therefore the users on the client machine need not interact during the install.



ATTENTION

You must perform the client installation for each computer which needs to run the software application.

To install EBR software packages over a network

Step	Action
1	Insert the EBR DVD into the DVD drive. If the DVD does not autorun, use Windows Explorer to go to the <i>CDBROWSE.exe</i> file and then double-click the file to launch the program. The following Honeywell - EBR menu appears. 
2	Select Experion Backup/Restore Software Packages to continue. The Welcome dialog box appears.
3	Click Next. The End User License Agreement dialog box appears.
4	Click Next. The Third-Party Software Compatibility Policy dialog box appears.
5	Click Next. The User Information dialog box appears.

Step	Action
6	Type the requested information in the appropriate text box. Note: The License No. and Authorization No. are available from the License Authorization of Honeywell. This information determines which software packages you are authorized to install.
7	Click Next . The Package Selection dialog box appears.
8	For Manager, select the following package: - EBR Manager For Sever, select the following package: - EBR Server For Desktop, select the following package: - EBR Desktop
9	Click Install Package to continue. The Installable Packages dialog box appears.
10	a) Select Admin (Administrative Installation) option button as the installable component. b) Browse or enter the path to the new Distribution Point . For Manager, select the following path. C:\SoftwareDistribution\EBR_Manager For Server, select the following path. C:\SoftwareDistribution\EBR_Server For Desktop, select the following path. C:\SoftwareDistribution\EBR_Desktop Note: The selected folder must be empty. c) Accept the default Drive or select one from the list.

Step	Action
11	Click OK .
	Stop The task is complete. Return to the checklist that led to this task.

To publish the EBR software using Group Policy Object

Step	Action
1	Log on to a server machine as Domain Administrator .
2	Choose Start>Programs>Administrative Tools>Active Directory Users And Computers .
3	Right-click the desired Console , and then select Properties . For this example, the name of the Console is UISTEST.local.
4	Click the Group Policy tab.
5	Select Administrator Policy , and then click Properties . Note: You can create new groups and modify them (recommended).
6	Click the Security tab. Ensure the EBR Administrators group is given only Read and Apply Group Policy permission.
7	Click OK . Note: Do not use the Deny option, because it can cause unwanted results.
8	Click Edit on the Group Policy tab.
9	On Group Policy Object Editor page browse to User Configuration>Software Settings>Software installation .
10	Right-click Software installation , and then select New>Package....

Step	Action
11	For Manager Select C:\SoftwareDistribution\EBR_Manager For Server Select C:\SoftwareDistribution\EBR_Server For Desktop Select C:\SoftwareDistribution\EBR_Desktop
	ATTENTION Verify the cursor is pointing to the Software Distribution point corresponding to the network and not to the local drive on the Server.
12	Create an install.zap file using notepad, as shown: [Application] FriendlyName = "EBR Desktop Media" SetupCommand = "\\TPS_SIT\SoftwareDistribution\EBR_Desktop\install.exe" DisplayVersion = 311
13	For Manager: Save the install.zap file in SoftwareDistribution>EBR_Manager. For Server: Save the install.zap file in SoftwareDistribution>EBR_Server. For Desktop: Save the install.zap file in SoftwareDistribution>EBR_Desktop.
14	Select Files of type as *.zap, and then select install.zap.
15	Click Open .
16	In the Deploy Software dialog box, click Publish , and then click OK .
	Stop The task is complete. Return to the checklist that led to this task.

To install EBR on clients using published software on Server

Step	Action
1	Log on to a Client machine as Domain Administrator .
2	Click Start>Control Panel>Add or Remove Programs . The Add or Remove dialog box appears.
3	Double-click Add New Programs . The client names appear in the display panel.
4	For Desktop: Select EBR Desktop Media . For Manager: Select EBR Manager Media . For Server: Select EBR Server Media .
5	Click Add . Note: You must restart the computer for EBR Server and Desktop after the installation is complete.
	Stop The task is complete. Return to the checklist that led to this task.

2.5 Installing the EBR Manager software package

The EBR Manager software package is designed for installation on the Manager Node in a centrally-managed environment. For a typical installation, the recommended package is:

- EBR Manager

The Manager Node must meet the following minimum requirements:

- Licensed authorization from Honeywell
- Dedicated server machine
- Windows Server 2003 Standard operating system
- Microsoft .NET Framework version 2.0
- The following free hard disk space:

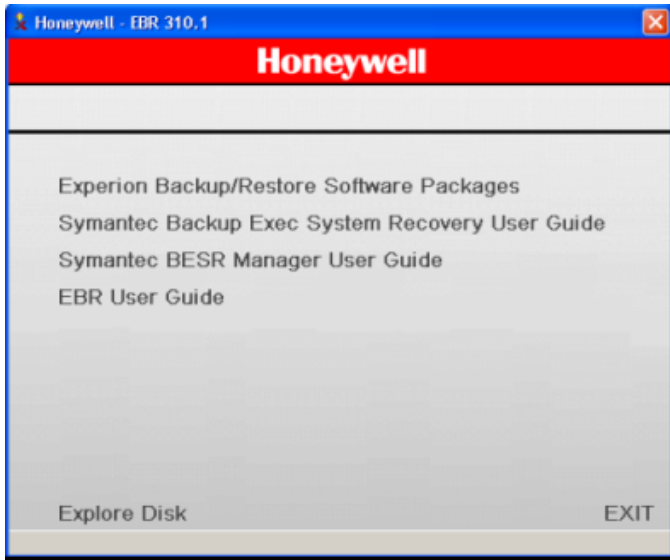
EBR software component	Required hard disk space
EBR Manager	1183 MB



ATTENTION

- You must complete this task for the Manager Node.
 - The computer used as the Backup Repository must meet the following requirements:
 - It must *not* be an Experion server, PHD server, or APP node. Honeywell recommends installing the Manager on a dedicated server that, in addition, operates as the Backup Repository, provided it is *not* a domain controller.
 - It must contain enough disk space to store all the intended backup files.
-

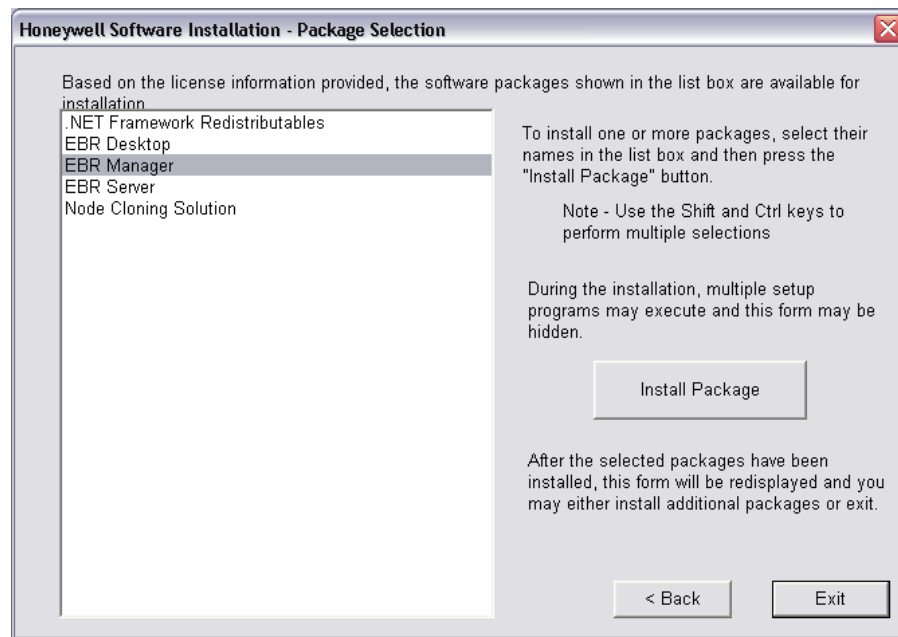
To install the EBR Manager software package

Step	Action
	ATTENTION Before attempting this installation, ensure that you are logged on to the node as a local administrator .
1	Insert the EBR DVD into the DVD drive. If the DVD does not autorun, use Windows Explorer to go to the <i>CDBROWSE.exe</i> file, and then double-click the file to launch the program. The following Honeywell - EBR menu appears. 
2	Select Experion Backup/Restore Software Packages to continue. The Welcome dialog box appears.
3	Click Next. The End User License Agreement dialog box appears.

2. Installing and configuring EBR

2.5 Installing the EBR Manager software package

Step	Action
4	Click Next . The Third-Party Software Compatibility Policy dialog box appears.
5	Click Next . The User Information dialog box appears.
6	Type the requested information in the appropriate text box. Note: The License No. and Authorization No. are available from the License Authorization of Honeywell. This information determines which software packages you are authorized to install.
7	Click Next . The following Package Selection dialog box appears.



2. Installing and configuring EBR

2.5 Installing the EBR Manager software package

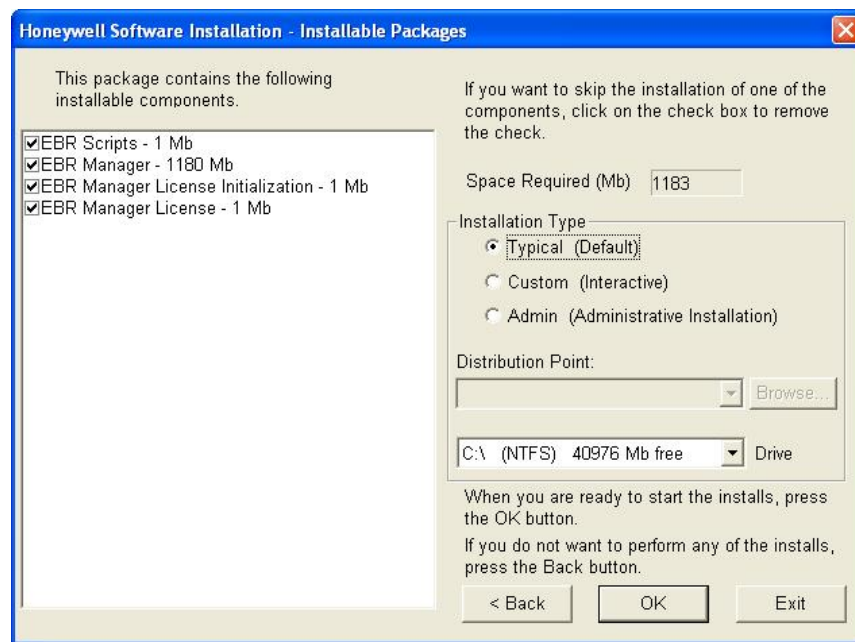
Step	Action
------	--------

- | | |
|---|---|
| 8 | Select the following software package from the list: <ul style="list-style-type: none">– EBR Manager |
|---|---|

Note: Unless you know that .Net Framework 2.0 or 3.0 is already installed on this node, select **.Net Framework Redistributables**.

- | | |
|---|---|
| 9 | Click Install Package to continue. |
|---|---|

The following **Installable Packages** dialog box appears.



- | | |
|----|---|
| 10 | a) Accept the default installable components, and then select the Typical (Default) option button. |
| | b) Accept the default Drive or select one from the list. |



ATTENTION

Do not select the **Custom** or **Admin** options unless advised by Honeywell.

2. Installing and configuring EBR

2.5 Installing the EBR Manager software package

Step	Action
11	Click OK to install the one or more software packages. Note: The software installation takes several minutes. The Symantec Backup Exec Manager License Agreement dialog box appears.
12	Select the I accept the terms... option.
13	Click Next. The Setup Type dialog box appears.
14	Select Complete under the Setup Type.
15	Click Next. The Logon Information dialog box appears. Note: EBR Manager asks for Logon Information only when you install it on Windows Server 2003 Standard operating system.

The screenshot shows a Windows-style dialog box titled "Symantec Backup Exec System Recovery Manager". Inside, the "Logon Information" section is active, with the instruction "Specify a user name and password". Below this, there is a text field for "User name:" containing "EBRManager", and two text fields for "Password:" and "Confirm password:", both containing "*****". The Symantec logo is in the top right corner. At the bottom, there are three buttons: "< Back", "Next >", and "Cancel". The "Next >" button is highlighted. The "InstallShield" logo is visible in the bottom left corner of the dialog area.

Step	Action
	ATTENTION • Use the provided username (EBRManager) and password (Manager) to create System Admin account during EBR Manager installation. It is mandatory for activating the EBR Manager License. • You can recreate System Account after installation and activation of the license.
16	Type EBRManager (username) and Manager (Password) to create a System Admin account. Note: The password cannot contain the following special characters: ('- apostrophe), ("quotation mark), or (\ backslash). When you logon as <i>System Admin</i>, you can set up other role-based user accounts with varying levels of privileges.
	ATTENTION The log-on password is required for the Manager to access the Manager database—it is <i>not</i> associated with any Microsoft Windows account. Record the log-on password and store it in a secure location. Symantec has no method of restoring a password to the server. If the password is lost, the only recovery method is to uninstall and then reinstall the EBR Manager software.
17	Click Next . The Ready to Install the program dialog box appears.
18	Click Install to continue. Note: The software installation takes several minutes to complete. After successful installation, the InstallShield Wizard Completed dialog box appears.
19	Ensure the Launch Symantec Backup Exec System Recovery Manager Console option is selected. Click Finish to close the installation wizard. The Symantec Backup Exec System Recovery Manager dialog box appears.

Step	Action
	ATTENTION The EBR Manager Console takes a few minutes to invoke. Do not close installation Main Menu. EBR License Activation proceeds after EBR Manager Console invocation and closing.
20	Enter the password.
21	Click OK . The message Logging on to the Backup Exec System Recovery Manager for first time appears.
22	Click OK . The Security Warning dialog box appears.
23	Click Yes to install the certificate. The EBR Manager Console is invoked.
24	Close the Console.
	ATTENTION You must close the EBR Manager Console for activating the Manager License.
25	When the EBR Manager License is activated, the EBR Manager Licensing dialog box appears.
	
26	Click OK .
	Stop The task is complete. Return to the checklist that led to this task.

To Configure the Windows Firewall

Step	Action
1	Choose Start > Control Panel > Windows Firewall .
2	On the General tab, select the On (recommended) option button.
3	Click the Exceptions tab.
4	Scroll through the Programs and Services list and look for the DCOM TCP 135 and Recovery Management names.
5	If DCOM TCP 135 is not listed, click Add Port , and then perform the following : a) In the Name box, type "DCOM TCP 135". b) In the Port number box, type "135". c) Select the TCP option button, and then click OK.
6	If Recovery Management is not listed, click Add Port , and then perform the following: a) In the Name box, type "Recovery Management". b) In the Port number box, type "8443". c) Select the TCP option button, and then click OK.
7	Click OK to close the Windows Firewall dialog box.
	Stop The task is complete. Return to the checklist that led to this task.

To create packages for Easy Installer

An Easy Installer is a single installation that contains one or more packages that you have previously created. It resides on the EBR Manager.

The EBR Manager contains the following pre-installed packages.

- Backup Exec System Recovery
- Management Control
- Indexing Server (Optional)

Perform the following steps to create packages for Easy Installer.

Step	Action
1	On the Tools menu, click Create Easy Installer . The Create Easy Installer dialog box appears.
2	In Choose Install Packages page, select the Management Control package. Note: Ensure you have already installed EBR Server or EBR Desktop on the Client node using the EBR DVD.
3	Click Next . The Easy Installer Client Option appears.
4	Accept the default options. Click Next . The Easy Installer Options page appears.
5	Accept the default selection. Click Next .
6	In the Easy Installer Options page, click Next .
7	In Completing the Easy Installer page, click Finish .



ATTENTION

Indexing Server is an optional feature. EBR Manager does not depend on Indexing service for central management of the Backup and Restore system.

The Indexing service must be installed on Windows Server 2003 using an EasyInstaller. It enables the Backup Exec Retrieve utility to maintain a folder of files that allow users to search file names, file versions, and recent activity for recovery by way of the Backup Exec Retrieve.

Indexing Service is a memory-intensive process. Each Indexing Server process consumes additional memory as it is based on the number of file versions that are indexed. Therefore, Indexing Server must not be installed (using EasyInstaller) on Honeywell system nodes. It can be used on Non-Honeywell Nodes.

For more information on Indexing Server and Backup Exec Retrieve, refer *Symantec Backup Exec™ System Recovery Manager Implementation Guide*.

To download Easy Installer on a client

You can add computers to the Manager Console by installing Management Control package using an Easy Installer. The Management Control component publishes information back to the server, and populates the EBR Manager Console.

Prerequisite

You must note the DownloadCenter URL from the EBR Manager. Then, browse to Backup Exec System Recovery Download Center Web page on the Client node.

Step	Action
1	Type the DownloadCenter URL in the web browser on the client node as follows: <i>https://<server_name or IP_address>:8443/DownloadCenter</i>
2	Type the user name (<i>EBRManager</i>) and password (<i>Manager</i>). The download center screen appears.

Step	Action
3	For server, Click Management Control(8.0.1.24979) For Desktop, Click Management Control(8.0.1.24979) The File Download dialog box appears.
4	Click Save and browse to a location on the local drive to save the install packages.
5	Select the file and run it.
6	Restart the node when it prompts to do so.



REFERENCE – EXTERNAL

For more information about creating packages using Easy Installers and adding computers to the console, see the following documentation on the EBR product DVD: Chapter 5, “Configuring Backup Exec System Manager” in the *Symantec Backup Exec™ System Recovery Manager Implementation Guide*.



ATTENTION

You must activate the License for EBR Desktop/ Server (Symantec Backup Exec System Recovery software) using the EBR R311.1 DVD before installing Management Control Software on Desktop / Server to add them to the Management Console.

2.6 Installing the EBR Server software package

The EBR Server software package is designed for installation on a Server Node in a locally-managed environment. For a typical installation, the recommended package is:

- EBR Server

The Server Node must meet the following minimum requirements:

- Licensed authorization from Honeywell
- Windows Server 2003 Standard operating system
- Microsoft .NET Framework version 2.0 (although version 2.0 is available, it does not replace version 1.1, but both can be installed on the same Server Node)

The following free hard disk space:

EBR software component	Required hard disk space
Server	582 MB

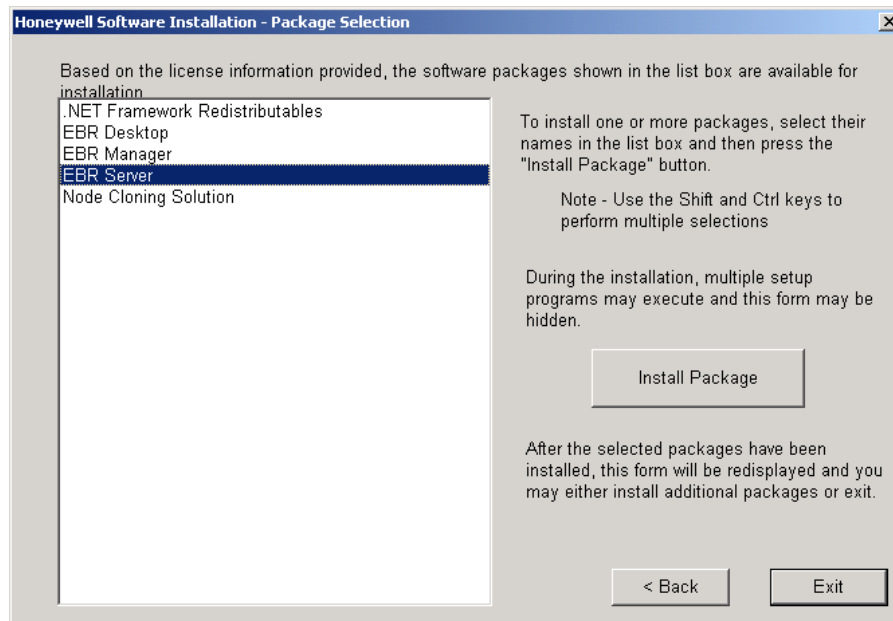
**ATTENTION**

- Before installing the EBR software, close all applications, such as Control Builder and Configuration Studio.
- Restart the computer after installing the EBR software.

To install the EBR Server software package

Step	Action
 ATTENTION	Before installing, ensure that you are logged on to the node as a local administrator .
1	Insert the EBR DVD into the DVD drive. If the DVD does not autorun, use Windows Explorer to go to the <i>CDBROWSE.exe</i> file, and then double-click the file to launch the program. The following Honeywell - EBR menu appears. 
2	Select Experion Backup/Restore Software Packages to continue. The Welcome dialog box appears.
3	Click Next. The End User License Agreement dialog box appears.

Step	Action
4	Click Next. The Third-Party Software Compatibility Policy dialog box appears.
5	Click Next. The User Information dialog box appears.
6	Type the requested information in the appropriate text box. Note: The License No. and Authorization No. are available from the License Authorization of Honeywell. This information determines which software packages you are authorized to install.
7	Click Next. The following Package Selection dialog box appears.



- 8 Select **EBR Server** from the list.
- Note:** Unless you know that .Net Framework 2.0 or 3.0 is already installed on this node, select .Net Framework Redistributables.

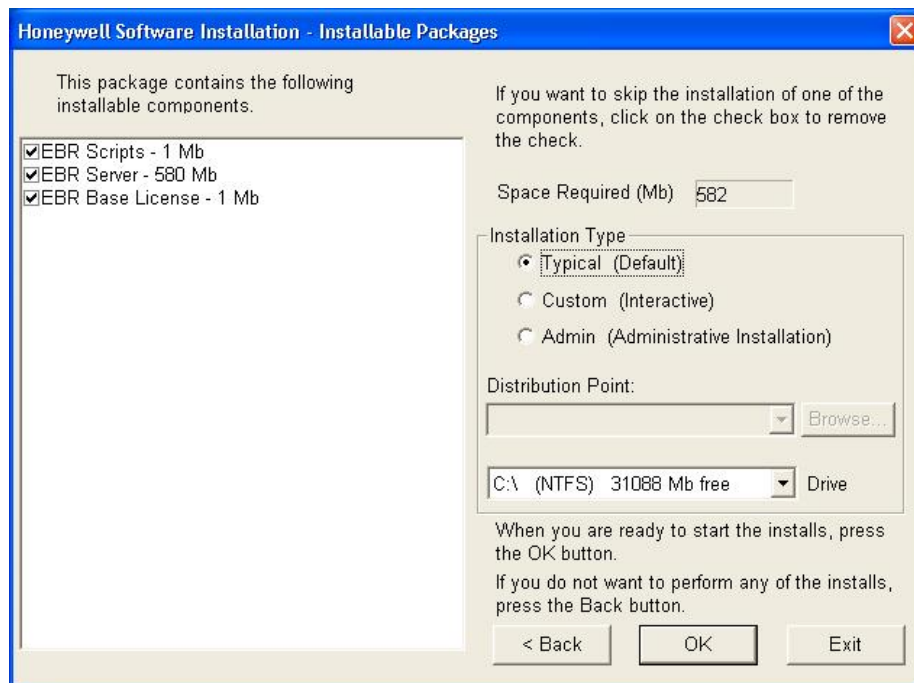
2. Installing and configuring EBR

2.6 Installing the EBR Server software package

Step	Action
------	--------

- | | |
|---|---|
| 9 | Click Install Package to continue. |
|---|---|

The following **Installable Packages** dialog box appears.



- | | |
|----|---|
| 10 | d) Accept the default installable components in the list, and then select the Typical (Default) option button. |
| | e) Accept the default Drive or select one from the list. |



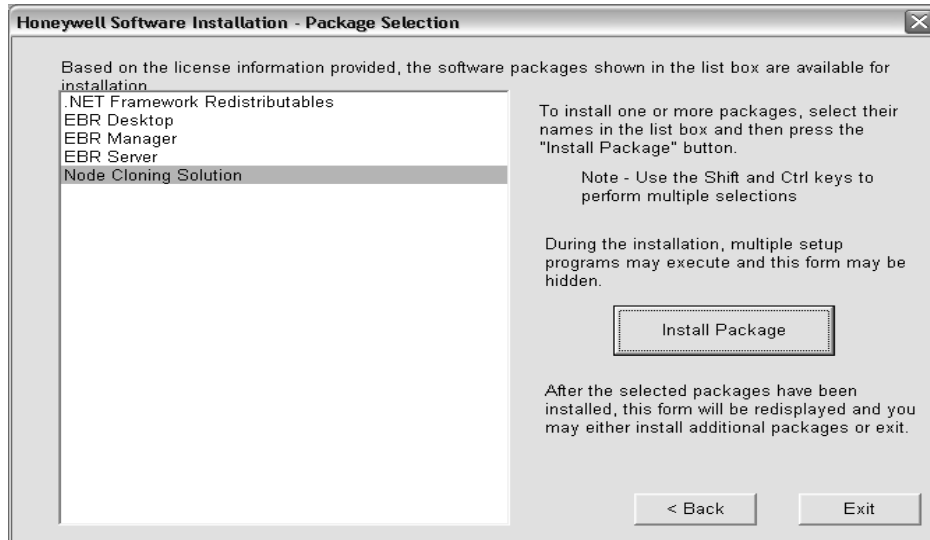
ATTENTION

Do not select the **Custom** or **Admin** options unless advised by Honeywell.

- | | |
|----|---|
| 11 | Click OK to install the one or more software packages. |
|----|---|

The **EBRLicenseAuto** dialog box appears.

Step	Action
	ATTENTION Installation of Node Cloning Solution package is optional. Install this package only if you want to clone nodes. This package installs scripts supporting cloning procedure. <i>Perform steps 12 through 17 to install Node Cloning Solution (this package is used for loading multiple Stations from a common image)</i>
12	The Package Selection dialog box appears.

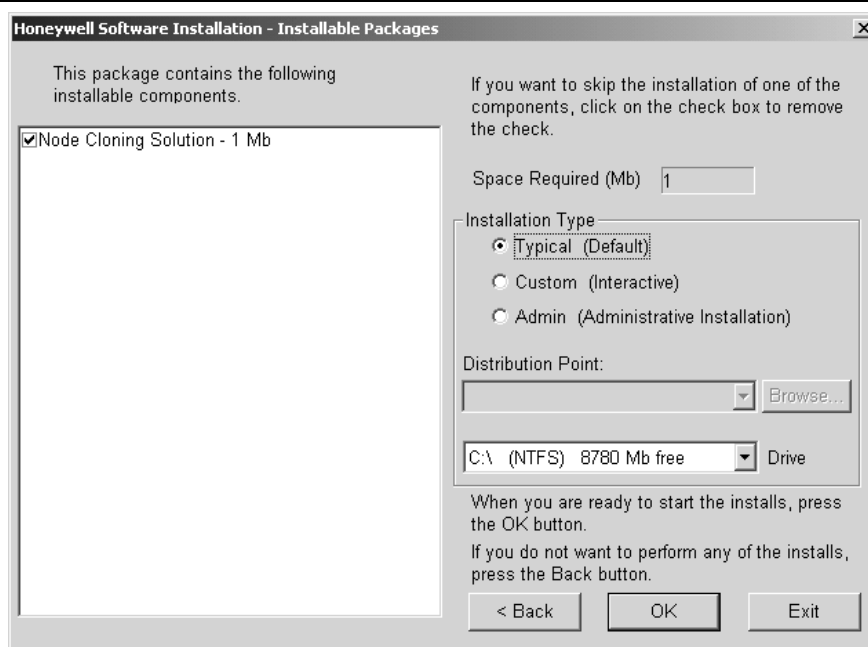


- 13 Select **Node Cloning Solution** from the list.
- 14 Click **Install package**.
- The **Installable Package** dialog box appears.
-

2. Installing and configuring EBR

2.6 Installing the EBR Server software package

Step	Action
------	--------



- 15 a) Accept the default installable components in the list, and then select the **Typical (Default)** option button.
- b) Accept the default **Drive** or select one from the list.



ATTENTION

Do not select the Custom or Admin options unless advised by Honeywell.

- 16 Click **OK** to install the one or more software packages.

The Package Selection dialog box appears.

Step	Action
17	Click Exit . The Are you sure you want to exit? Message box appears.
18	Click Yes . Note: Remove the EBR DVD from the DVD drive and restart Windows after the installation is complete.
	Stop The task is complete. Return to the checklist that led to this task.

2.7 Installing the EBR Desktop software package

The EBR Desktop software package is designed for installation on a Desktop Node in a locally-managed environment. For a typical installation, the recommended packages are:

- EBR Desktop

The Desktop Node must meet the following minimum requirements:

- Licensed authorization from Honeywell
- Windows XP Professional (SP2 or later) operating system
- Microsoft .NET Framework version 2.0 (although version 2.0 is available, it does not replace version 1.1, but both can be installed on the same Workstation Node)

The following free hard disk space:

EBR software component	Required hard disk space
Desktop	581 MB



ATTENTION

- Before installing the EBR software, close all applications such as Control Builder and Configuration Studio.
 - You must restart the computer after installing the EBR software.
-

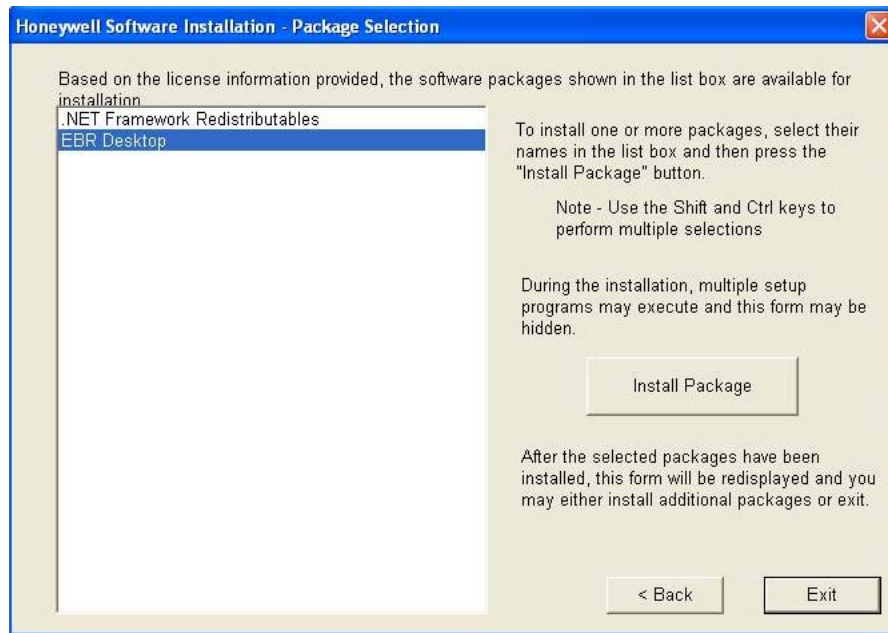
To install the EBR Desktop software package

Step	Action
	ATTENTION Before installing, ensure that you are logged on to the node as a local administrator .
1	Insert the EBR DVD into the DVD drive. If the DVD does not autorun, use Windows Explorer to go to the <i>CDBROWSE.exe</i> file, and then double-click the file to launch the program. The following Honeywell - EBR menu appears. 
2	Select Experion Backup/Restore Software Packages to continue. The Welcome dialog box appears.
3	Click Next. The End User License Agreement dialog box appears.

2. Installing and configuring EBR

2.7 Installing the EBR Desktop software package

Step	Action
4	Click Next. The Third-Party Software Compatibility Policy dialog box appears.
5	Click Next. The User Information dialog box appears.
6	Type the requested information in the appropriate text box. Note: The License No. and Authorization No. are available from the License Authorization of Honeywell. This information determines which software packages you are authorized to install.
7	Click Next. The following Package Selection dialog box appears.

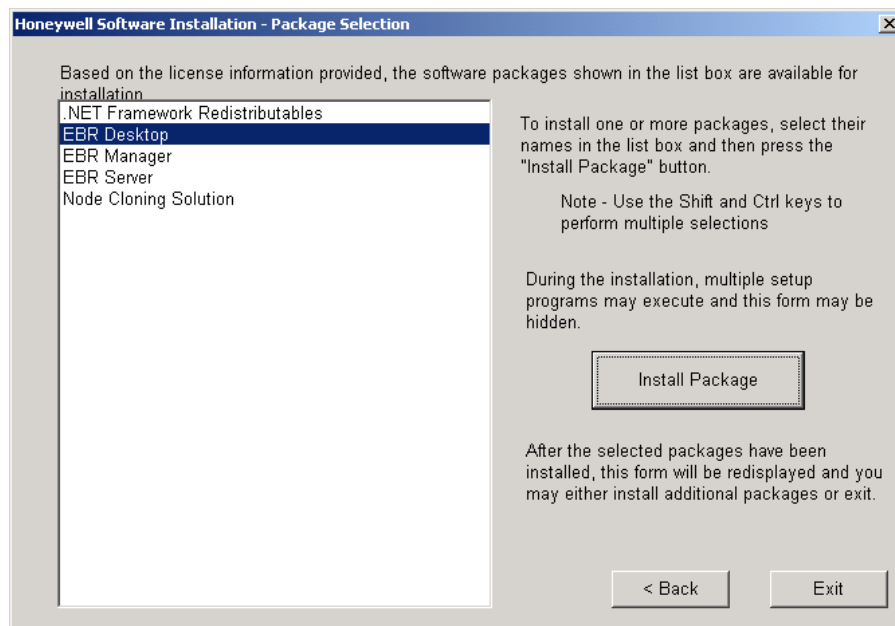


- 8 Select **EBR Desktop** from the list.
- Note:** Unless you know that .Net Framework 2.0 or 3.0 is already installed on this node, select .Net Framework Redistributables.

Step	Action
------	--------

- 9 Click **Install Package** to continue.

The following **Installable Packages** dialog box appears.



- 10 a) Accept the default installable components in the list, and then select the **Typical (Default)** option button.
b) Accept the default **Drive** or select one from the list.



ATTENTION

Do not select the **Custom** or **Admin** options unless advised by Honeywell.

- 11 Click **OK** to install the one or more software packages.

The **EBRLicenseAuto** dialog box appears.



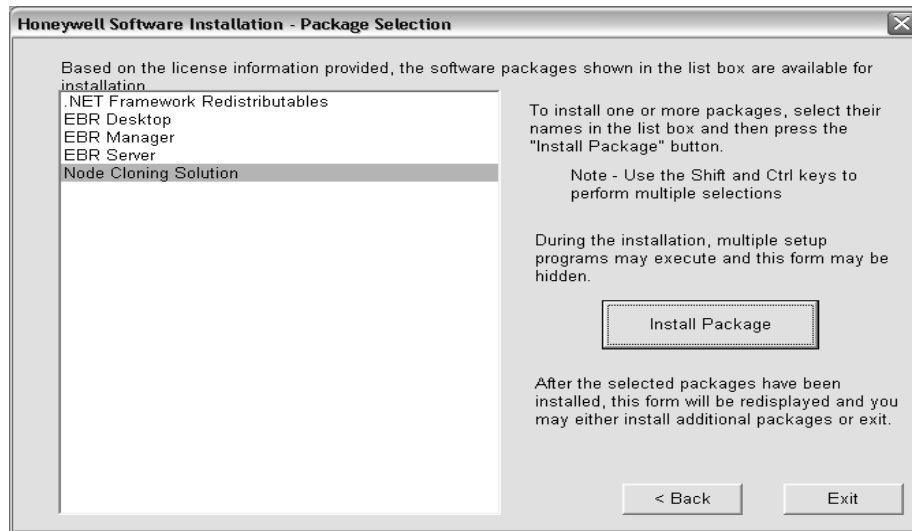
ATTENTION

Installation of Node Cloning Solution package is optional. Install this package only if you want to clone nodes. This package installs scripts supporting cloning procedure.

Step	Action
------	--------

Perform steps 12 through 17 to install Node Cloning Solution (this package is used for loading multiple Stations from a common image)

- 12** The **Package Selection** dialog box appears.



- 13** Select **Node Cloning Solution** from the list.

- 14** Click **Install package**.

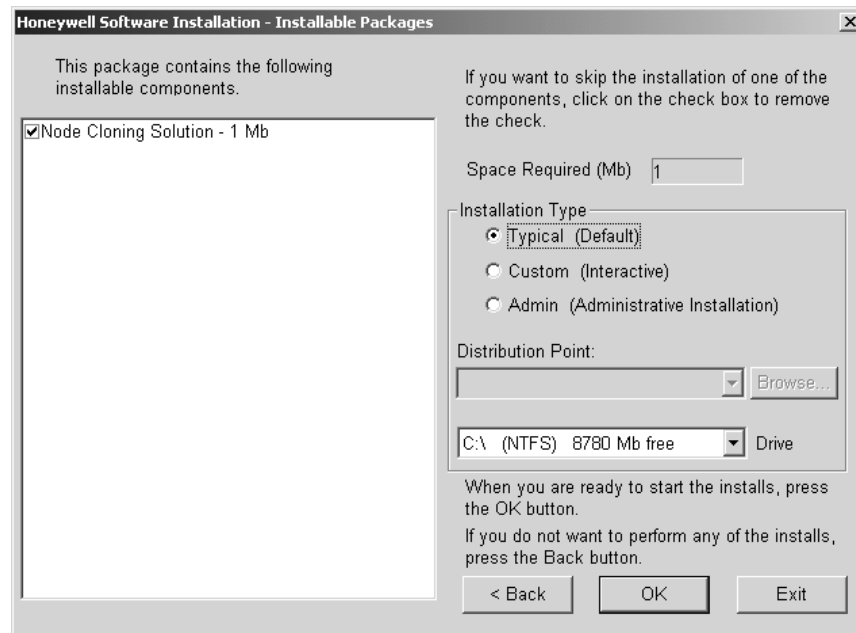
The **Installable Package** dialog box appears.

2. Installing and configuring EBR

2.7 Installing the EBR Desktop software package

Step

Action



- 15 a) Accept the default installable components in the list, and then select the **Typical (Default)** option button.
- b) Accept the default **Drive** or select one from the list.



ATTENTION

Do not select the Custom or Admin options unless advised by Honeywell.

- 16 Click **OK** to install the one or more software packages.
- The Package Selection dialog box appears.
-

2. Installing and configuring EBR

2.7 Installing the EBR Desktop software package

Step	Action
17	Click Exit . The Are you sure you want to exit? Message box appears.
18	Click Yes . Note: Remove the EBR DVD from the DVD drive and restart Windows after the installation is complete.
	Stop The task is complete. Return to the checklist that led to this task.

2.8 Configuring EBR Manager Console

The EBR Manager Console is typically installed on the Server Node, and is used for managing backups and deploying agents to Target Nodes in a centrally-managed environment.

You can use the EBR Manager Console to set backup defaults and notifications, add or remove nodes, and change the console password. In addition, you can cluster nodes within a group, which can be a set of nodes in the same operational area, a set of similar nodes, an Experion cluster, or some other logical arrangement in your plant.



ATTENTION

The EBR Manager menu lists a LiveUpdate option. This option is Not supported by Honeywell. The EBR software includes only versions that are validated by Honeywell for use in the control environment.

Starting the Manager

To start the Manager

- Double-click the **Symantec Backup Exec System Recovery** icon on the Windows desktop or notification area.

Or

- Click **Start > Programs > Symantec > Backup Exec System Recovery > Symantec Backup Exec System Recovery Manager**.

To set backup defaults

Step	Action
1	On the Manage page, click Destination , under Create . The Create Backup Destination dialog box appears.
2	Type the default location for backup images or click Browse to search.
CAUTION Do not use the hard disk of a Target Node to store backup image files. The performance of the node can be degraded if the hard disk is full.	
3	Type the user name and password in Network Credentials , and then click OK .
4	Use one of the following methods to set the default backup operation speed. For a Computer Group: a) On the Groups tab, select one or more group names in the table. b) Right-click a computer group and click Client Options . The Client Options dialog box appears.
5	On the left pane, click Performance .
6	Adjust the operation speed to 50 (recommended value).
7	Click OK .
 Stop The task is complete. Return to the checklist that led to this task.	

Enhancing Backup Performance

The Performance option (Choose **Tasks > Option > Settings**) lets you specify a default speed for backup or recovery processes. Moving the slider closer to Fast increases the speed at which the program backs up or recovers your computer. However, choosing a slower speed could improve the performance of your computer, especially if you are working on your computer during a backup or recovery.

Note: During a backup or recovery, you have the option to override this default setting to fit your needs at the time.

In addition, you can configure network throttling to limit the effects of backups on network performance.

To enable network throttling

Step	Action
1	On the menu bar, click Tasks > Options . The Options dialog box appears.
2	Click Performance .
3	Select Enable network throttling .
4	In the Maximum network throttling , enter the maximum amount (in KB) of network throughput that Backup Exec System Recovery can send per second.

Copying command files to PHD Nodes and Experion Servers with EBR installed

Installation of the EBR Server software package automatically copies command files to the *C:\HWIAC\backup* folder. After the installation, you must copy the command files to a specified Symantec folder in each Target Node, including those for which you deployed the EBR agent. To do this, you can copy the files to a shared folder on a network drive and then use Windows Explorer to paste them to the Symantec folder, or, you can use removable media to copy and paste the files.

To copy command files to a PHD Node with EBR installed

Step	Action
1	On the PHD Node, go to the <i>C:\HWIAC\backup</i> folder, and then copy the <i>phd_aftercapture.cmd</i> , <i>phd_beforecapture.cmd</i> and <i>delay.exe</i> files.
2	Paste the command file(s) in the following folder on that node. <i>C:\Documents and Settings\All Users\Application Data\Symantec\Backup Exec System Recovery\CommandFiles</i>
	Stop The task is complete. Return to the checklist that led to this task.

To copy command files to an Experion Server with EBR installed

Step	Action
1	On the Experion Server, go to the <i>C:\HWIAC\backup</i> folder, and then copy the <i>esv_db_beforecapture.cmd</i> file.
2	Paste the command file in the following folder on that server: <i>C:\Documents and Settings\All Users\Application Data\Symantec\Backup Exec System Recovery\CommandFiles</i>
	Stop The task is complete. Return to the checklist that led to this task.

Adding node groups and nodes

When you add a node to the Console, the EBR delivery agent is deployed to that node to establish a communication link for deployment of options and software packages. If the node is in a group, you must first add the node group and then add that node.

To add a node group

Step	Action
1	On the Manage Page, click Group.... The Create Group dialog box appears.
2	Type the group name and description in the text boxes, and then click OK .
	Stop The task is complete. Return to the checklist that led to this task.
	REFERENCE - EXTERNAL For more information, see <i>About organizing groups of computers > Adding groups to the Console</i> in the Symantec Backup Exec System Recovery Manager online help.

To add a node

Step	Action
1	On the Manage Page, click Assign Computers to Groups . The Assign Computers to Groups dialog box appears.
2	Select the computers that you want to assign to the group.
3	Click OK .
	Stop The task is complete. Return to the checklist that led to this task.
	REFERENCE - EXTERNAL For more information, see <i>About organizing groups of computers > Assigning computers to a group</i> in the Symantec Backup Exec System Recovery online help.

Removing node groups and nodes

To remove a node group and nodes



REFERENCE - EXTERNAL

For details about removing a node group, see *About organizing groups of computers > Unassigning computers from a group and Removing computers from a console* in the Symantec Backup Exec System Recovery online help.

Changing the Manager log-on password



ATTENTION

The console log-on password is required for the Manager to access the EBR Manager database—it is *not* associated with any Microsoft Windows account.

Record the console log-on password and store it in a secure location. Symantec does not provide a solution to restore a password to the server. If the password is lost, the only recovery method is to uninstall, and then reinstall the EBR Manager software.

To change the console log on password

Step	Action
1	Confirm that you are logged on to the EBR Manager.
2	Click Tools > Change Password .
3	Type the old and new passwords (case-sensitive) in the appropriate text boxes, and then click OK .

2.9 Configuring Desktop and Server User Interfaces

The Desktop User Interface on a Desktop node is used for managing backup jobs on that node in a locally-managed environment where EBR Manager is not used.

The Server User Interface on a Server node is used for managing backup jobs on that node in a locally-managed environment where EBR Manager is not used.

The Desktop User Interface is installed on a computer running the Windows XP Professional operating system.

The Server User Interface is installed on a computer running Windows Server 2003 Standard.

Starting the Desktop or Server User Interface

To start the Desktop or Server User Interface, double-click the Symantec Backup Exec Recovery 8 icon on the Windows desktop or notification area. Or, choose **Start > Programs > Symantec > Backup Exec System Recovery**.

To set backup defaults

Step	Action
1	Click Task > Options . The Options dialog box appears.
2	Under Settings , click General . a) Select the default location for the recovery points. b) Type the user name and password for Network Credentials to copy backup to a network drive.
CAUTION Do not use the hard disk of a Target Node to store backup image files. The performance of the node can be degraded if hard disk is full.	
3	- Click Performance and set the Default Performance slider to midpoint (recommended). - Select Enable network throttling to limit the impact of a backup on a network performance. - In the Maximum network throttling, enter the maximum amount (in KB) of network throughput that Backup Exec System Recovery can send per second.

Step	Action
4	Under Notifications, select the Log File. Browse <i>C:\Documents and Settings\All Users\Application Data\Symantec\Backup Exec System Recovery\Logs\</i>.
	ATTENTION Honeywell does <i>not</i> recommend using the SMTP e-mail notification option.
5	Click OK.
	Stop The task is complete. Return to the checklist that led to this task.
	REFERENCE - EXTERNAL For more details about setting default backup options, see <i>Getting Started > Configuring Backup Exec System Recovery default options > Selecting a default backup destination</i> in the Symantec Backup Exec System Recovery online help.

2.10 Configuring the Windows Firewall

Nodes with Experion software installed

If you installed EBR software on a node with Windows Server 2003 Standard SP1 or later, Windows XP Professional SP2 or later, *and* Experion software, you must enable and configure the Windows Firewall.



ATTENTION

You must complete this task on all nodes with EBR software *and* Experion software installed.

To configure Windows Firewall for nodes with Experion software installed

Step	Action
1	Click Start > Settings > Control Panel > Windows Firewall .
2	Click the Exceptions tab.
3	Click Add Program , and then click Browse to go to the following folder: C:\Program Files\Symantec\Backup Exec System Recovery\Agent\
4	Select the VProSvc.exe file, click Open , and then click OK .
5	Click Add Program , and then click Browse to go to the following folder: C:\Program Files\Symantec\Backup Exec System Recovery\Console
6	Select the VProConsole.exe file, click Open , and then click OK .
7	Click OK to close the Windows Firewall dialog box.
8	Close the Control Panel window.



Stop

The task is complete. Return to the checklist that led to this task.

Nodes without Experion software installed

If EBR software is installed on a computer that does not have Experion software installed, you must complete additional configuration steps for the Windows Firewall.



ATTENTION

You must complete this task on all nodes with EBR software installed but *without* Experion software.

To configure Windows Firewall for nodes without Experion software installed

Step	Action
1	Click Start > Settings > Control Panel > Windows Firewall .
2	On the General tab, select the On (recommended) option button.
3	Click the Exceptions tab.
4	In the Programs and Services list, make sure that File and Printer Sharing is selected. Look for the TCP 135 and Recovery Management names.
5	If TCP 135 is not listed, click Add Port , and then perform the following: a) In the Name box, type "TCP 135". b) In the Port number box, type "135". c) Select the TCP option button and then click OK.
6	If Recovery Management is not listed, click Add Port , and then perform the following: a) In the Name box, type "Recovery Management". b) In the Port number box, type "8443". c) Select the TCP option button, and then click OK.
7	Click OK to close the Windows Firewall dialog box.
8	Close the Control Panel window.



Stop

The task is complete. Return to the checklist that led to this task.

2.11 Integrating EBR events into Experion System Management Displays

To allow integration of EBR events into the Experion System Management Display, the `C:\HWIAC\filters` folder on each Experion Target Node must contain the *ebrevts.xml* file. This file is usually contained in the EBR Manager Node.

- For a Target Node with EBR installed locally, the *ebrevts.xml* file is automatically added to its `C:\HWIAC\filters` folder.
- For a Target Node with EBR deployed from the EBR Manager Console, the *ebrevts.xml* file must be manually added to the same location on that Target Node.

Use the following procedure to add the *ebrevts.xml* file to an Experion Target Node.

To integrate EBR events into the Experion System Management Display

Step	Action
1	On the EBR Manager Node, go to the <code>C:\HWIAC\backup</code> folder and then copy the <i>ebrevts.xml</i> file.
2	Paste the <i>ebrevts.xml</i> file into the <code>C:\HWIAC\filters</code> folder of each Experion Target Node. (You may need to create the <i>filter</i> folder.)

2.12 Validating Drivers

The driver validation utility determines if the EBR Recovery CD contains the appropriate drivers for data storage hardware and network hardware on the node being restored.



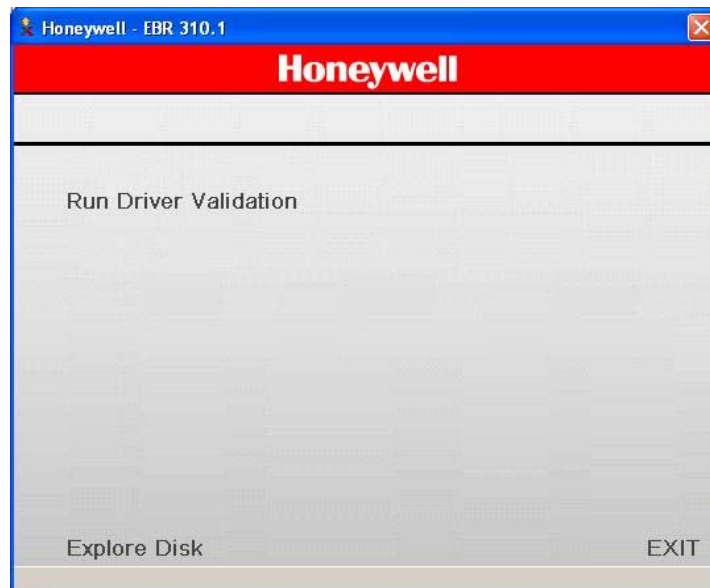
ATTENTION

The Honeywell FTE drivers are not available in the Honeywell recovery CD. The Driver Validation tool displays the same as not available.

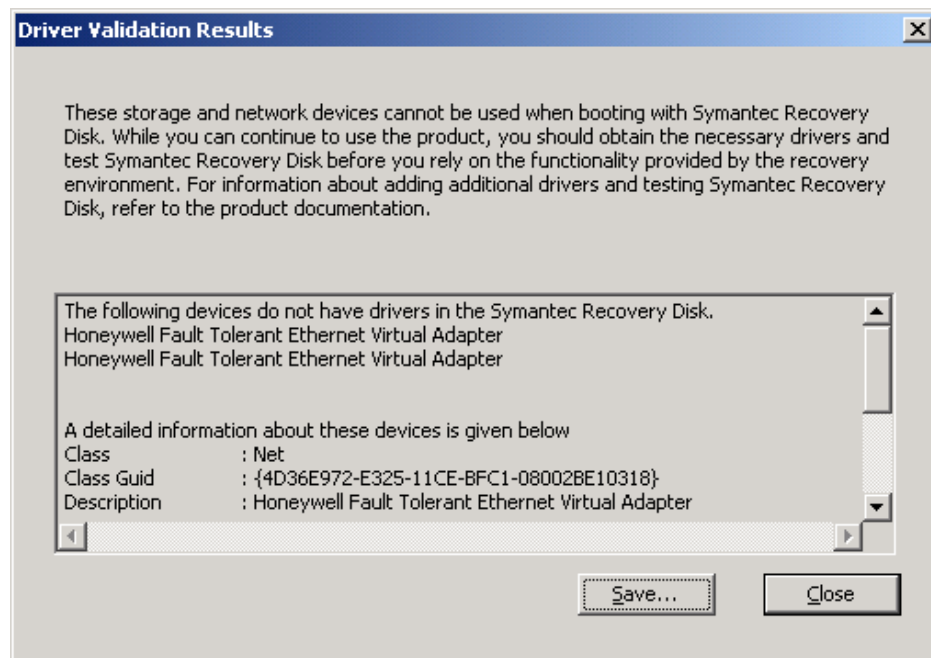
If a node has Honeywell FTE driver installed on it, you must install the FTE driver on the restored node again using the EBR Backup image.

To run the driver validation utility

Step	Action
1	Insert the EBR Recovery CD into the CD drive. If the CD does not autorun, use Windows Explorer to go to the <i>CDBROWSE.exe</i> file, and then double-click the file to launch the program. The following Honeywell – EBR menu appears.



Step	Action
2	Select Run Driver Validation to continue. The Driver Validation dialog box appears.
3	Click Validate Drivers . The following message appears. “All detected storage devices of network cards have drivers in the Symantec Recovery disk.” If driver validation fails then the following Driver Validation Results dialog box appears.



Step	Action
4	a) If no errors are reported, click Save to continue. b) If Honeywell Fault Tolerant Ethernet Virtual Adapter errors are reported, you can safely ignore them (such errors does not interfere with a recovery and are expected). Click Close to continue. c) If other errors are reported, you can create a Custom recovery CD. Click Close to continue. Note: To create a custom recovery CD, see section 2.13, "Creating a custom recovery CD."
5	Click OK to terminate the program.
6	On the Honeywell - EBR menu, click Exit , and then Yes .
7	Remove the EBR Recovery CD from the CD drive.

2.13 Creating a custom recovery CD

Even if driver validation succeeds and your Honeywell Custom Recovery CD appears to work, you must create a Custom Recovery CD. A custom CD contains your computer's current network and storage device drivers, ensuring that in an emergency you can get to the recovery points required to restore your computer.



ATTENTION

You must have a writeable DVD/CD-RW drive to create a Honeywell Custom Recovery CD.

To create a Custom Recovery CD

Step	Action
1	Start Backup Exec System Recovery .
2	Attach and turn on all storage devices and network devices that you want to make available.
3	Insert the writable (or blank) Honeywell Custom Recovery CD you are creating, into your CD-RW drive.
4	In the main Backup Exec System Recovery dialog box, click Tasks > Create Recovery Disk , and then click Next .
5	If prompted, click Browse , select the drive that contains the Honeywell Recovery CD, and then click OK .
6	Click Next . Perform one of the following: • When the exit wizard appears, click Finish. • Click Custom > Add. The Add Drivers dialog box appears. Select this option only if you know which drivers to select.
7	Follow the on-screen instructions to complete the wizard.

CAUTION

Ensure to test your new customized Honeywell Recovery CD to ensure that it can start your computer and that you can access the drive containing your recovery points.

Testing the Custom Recovery CD

You must test the Custom Recovery CD to ensure that the recovery environment runs properly on your computer.

Perform the following to test the Custom Recovery CD.

Step	Action
1	Run the driver validation tool to test whether the Honeywell custom recovery CD works with the network cards and storage devices on the computer.
2	Boot your computer using the custom recovery CD.

3. Removing EBR

3.1 Removing the EBR Manager software



ATTENTION

You must restart the computer, after removing the EBR software.

To remove the EBR Manager software

Step	Action
1	Choose Start > Control Panel > Administrative Tools > Services .
2	Double-click SymSnapService . The SymSnapService Properties dialog box appears.
3	Click the General tab and then perform the following: a) In Startup type , select Disabled , and then click Apply . b) In Service status , click Stop . c) Click OK to close the Properties dialog box.
4	Click Start > Control Panel > Add or Remove Programs .
5	Select EBR Scripts , and then click Remove . Click Yes to proceed.
6	Select EBR Manager License , and then click Remove . Click Yes to proceed.
7	Select Symantec Backup Exec System Recovery , and then click Remove . Click Yes to proceed.
8	Select LiveUpdate 3.2 (Symantec Corporation) , and then click Remove . Click Yes to proceed.
9	Select Symantec Backup Exec System Recovery Manager , and then click Remove . Click Yes to proceed.
10	In the Add or Remove Programs dialog box, click Yes to keep the data. Note: Even if you click No , some shared Symantec folders may remain in the <i>Program Files</i> directory.
11	When the You must restart your system... dialog box appears, click Yes .

3.2 Removing the EBR Server and Desktop software

**ATTENTION**

Removing the EBR software requires you to restart the computer.

To remove the EBR Desktop software

Step	Action
1	Click Start > Control Panel > Administrative Tools > Services .
2	Double-click SymSnapService . The SymSnapService Properties dialog box appears.
3	Click the General tab, and then perform the following: a) In Startup type, select Disabled, and then click Apply. b) In Service status, click Stop. c) Click OK to close the Properties dialog box.
4	Click Start > Settings > Control Panel > Add or Remove Programs .
5	Select LiveUpdate 3.2 (Symantec Corporation) , and then click Remove . Click Yes to proceed.
6	Select Ebrbase License , and then click Remove . Click Yes to proceed.
7	Select Symantec Backup Exec System Recovery , and then click Remove . Click Yes to proceed.
8	In the Add or Remove Programs dialog box, click Yes to keep the data. Note: Even if you click No , some shared Symantec folders may remain in the <i>Program Files</i> directory.
9	When the You must restart your system... dialog box appears, click Yes .

4. Restoring nodes and Experion databases

4.1 About this section

This section provides instructions for restoring various nodes and Experion Server databases after recovery from a failure, such as a hardware failure, virus infection, or database corruption.

Sections 4.2 and 4.3 describe node restoration based on whether the node is being restored to *different* computer hardware from the computer from which the backup was made, or to *substantially identical* computer hardware.

In addition, this section provides instructions for restoring the EMDB and ERDB databases for Experion Servers.



ATTENTION

Use the appropriate checklists that follow to restore nodes and databases after recovery from an unexpected event. These checklists specify the correct sequence of tasks for a successful restoration.

4.2 Checklists for restoring nodes with different hardware

To use the Restore AnyWhere option to restore a node to a computer with different hardware, the node must have Windows 2003 Server Standard or Windows XP Professional.

Restoring a non-redundant node with different hardware

Step	Task	Go to	Done?
1	Replace the failed node and reconnect related cables.	n/a	
2	Use EBR to restore the hard disk. Make sure to do the Restore AnyWhere steps.	page 85	
3	Configure a MAC address for new NIC cards.	page 88	
4	For an Experion Server, ensure to run the post-restore command.	page 92	

Restoring redundant Experion Server A with different hardware

Step	Task	Go to	Done?
1	Replace the failed node and reconnect related cables.	n/a	
2	Use EBR to restore the hard disk. Make sure to do the Restore AnyWhere steps.	page 85	
3	Use a Station to connect to the primary Experion Server, and then synchronize both Experion Servers.	page 100	

Note that the task performed to restore redundant Experion Server A needs to be repeated for redundant Experion Server B.

4.3 Checklists for restoring nodes with substantially identical hardware

Restoring a non-redundant node with substantially identical hardware

Step	Task	Go to	Done?
1	Replace the failed node and reconnect related cables.	n/a	
2	Use EBR to restore the hard disk.	page 85	

Restoring redundant Experion Server A with substantially identical hardware

Step	Task	Go to	Done?
1	Replace the failed node and reconnect related cables.	n/a	
2	Use EBR to restore the hard disk.	page 85	
3	Disable ERDB/EMDB replication for Experion Server B.	page 95	
4	Delete the contents of this folder on Experion Server A: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i>	n/a	
5	Restore the ERDB for Redundant Experion Server A.	page 96	
6	Restore the EMDB for Redundant Experion Server A.	page 98	
7	Use a Station to connect to the primary Experion Server, and then synchronize both Experion Servers.	page 100	

4. Restoring nodes and Experion databases
4.3 Checklists for restoring nodes with substantially identical hardware

Restoring redundant Experion Server B with substantially identical hardware

Step	Task	Go to	Done?
1	Replace the failed node and reconnect related cables.	n/a	
2	Use EBR to restore the hard disk.	page 85	
3	Disable ERDB/EMDB replication for Experion Server B.	page 95	
4	Delete the contents of this folder on Experion Server B: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i>	n/a	
5	Restore the ERDB for Experion Server B.	page 97	
6	Restore the EMDb for Experion Server B.	page 99	
7	Use a Station to connect to the primary Experion Server, and then synchronize both Experion Servers.	page 100	

4.4 Checklists for restoring nodes after a hard disk failure

Restoring a non-redundant node after a hard disk failure

Step	Task	Go to	Done?
1	Replace the hard disk and ensure the address pinning is identical to the original unit.	n/a	
2	Use EBR to restore the hard disk.	page 85	

Restoring redundant Experion Server A after a hard disk failure

Note that the task performed to restore redundant Experion Server A needs to be repeated for redundant Experion Server B.

Step	Task	Go to	Done?
1	Replace the hard disk and ensure the address pinning is identical to the original unit.	n/a	
2	Use EBR to restore the hard disk.	page 85	
3	Disable ERDB/EMDB replication for Experion Server B.	page 95	
4	Delete the contents of this folder on Experion Server A: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i> Note: Delete the contents of this folder on Experion Server B when Restoring redundant Experion Server B.	n/a	
5	Restore the ERDB for Redundant Experion Server B.	page 97	
6	Restore the EMDB for Redundant Experion Server B.	page 99	
7	Use a Station to connect to the primary Experion Server, and then synchronize both Experion Servers.	page 100	

4.5 Checklists for restoring nodes after recovery from a virus infection

Considerations

- Depending on your local Operations Policy about viruses, you may want to disconnect the infected nodes from all networks and restore them from removable media to avoid infecting the rest of the network.
- Recovery from a virus infection may require restoring an entire hard disk or set of hard disks, or it may simply require recovery of a few quarantined files.
- This procedure can confirm whether or not the backup image contains the virus, but it cannot remove the infected file from the backup image. If a backup image is infected, some uninfected files may still be safely recovered, but the backup image must not be used for a full disk recovery.



TIP

For this task, you do not need to reformat the hard disk because EBR restoration overwrites the entire hard disk.

4. Restoring nodes and Experion databases

4.5 Checklists for restoring nodes after recovery from a virus infection

Restoring a non-redundant node after recovery from a virus infection

Step	Task	Go to	Done?
1	Remove the computer from the network.	n/a	
2	Use the Recovery Point Browser to mount the intended backup image as a drive.	page 94	
3	Use the virus scan software to scan the drive specified in step 2.	Virus scan software documentation	
4	If the virus scan detects a virus in the backup image, use a previously archived, virus-free image to restore the hard disk.	n/a	
5	Use EBR to restore the hard disk.	page 85	
6	Add the computer back to the network.	n/a	

4. Restoring nodes and Experion databases
4.5 Checklists for restoring nodes after recovery from a virus infection

Restoring redundant Experion Server A after recovery from a virus infection

Step	Task	Go to	Done?
1	Remove the computer from the network.	n/a	
2	Reformat the hard disk.	Microsoft documentation	
3	Use EBR to restore the hard disk.	page 85	
4	Use the virus scan software to scan the selected disk drive.	Virus scan software documentation	
5	If the virus scan detects a virus in the backup image, use a previously archived backup image to restore the hard disk.	n/a	
6	Disable ERDB/EMDB replication for Experion Server B.	page 95	
7	Delete the contents of this folder on Experion Server A: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i>	n/a	
8	Restore the ERDB for Experion Server B.	page 97	
9	Restore the EMDb for Experion Server B.	page 99	
10	Use a Station to connect to the primary Experion Server, and then synchronize both Experion Servers.	page 100	

4. Restoring nodes and Experion databases

4.5 Checklists for restoring nodes after recovery from a virus infection

Restoring redundant Experion Server B after recovery from a virus infection

Step	Task	Go to	Done?
1	Remove the computer from the network.	n/a	
2	Reformat the hard disk.	Microsoft documentation	
3	Use EBR to restore the hard disk.	page 85	
4	Use your virus scan software to scan the selected disk drive.	Virus scan software documentation	
5	If the virus scan detects a virus in the backup image, use a previously archived backup image to restore the hard disk.	n/a	
6	Disable ERDB/EMDB replication for Experion Server B.	page 95	
7	Delete the contents of this folder on Experion Server B: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i>	n/a	
8	Restore the ERDB for Experion Server B.	page 97	
9	Restore the EMDb for Experion Server B.	page 99	
10	Use a Station to connect to the primary Experion Server, and then synchronize both Experion Servers.	page 100	

4.6 Checklists for restoring selected files from a backup

If you can start the node from its operating system, but have lost data or program files on the hard disk (excluding some operating system files), you can recover selected files or folders from a backup image without restoring the entire hard disk. This recovery method—known as a “hot restore”—is useful when node failures are traced to a specific file or folder, or important data files are lost and you do not want to use the Restore Drive Wizard to restore an entire backup image.

Considerations using EBR

- If your backup image is on removable media, copy the backup to a hard disk and then restore the files from the hard disk.
- If you are restoring from an incremental backup, the restore process automatically detects any previous incremental and the base backup image associated with the selected incremental.



ATTENTION

Recovery of selected files and folders does not recover the associated registry settings.

Restoring selected files to a node

Task	Go to	Done?
Use EBR to restore the selected file.	page 93	

4.7 Checklists for restoring corrupted Experion Server databases

If the ERDB or EMDB on a redundant Experion Server becomes corrupted, first try to restore the database from the uncorrupted Experion Server. If both databases are corrupted, you must restore them from a backup.

Restoring a corrupted ERDB or EMDB on a non-redundant server

Task	Go to	Done?
Use EBR to restore the corrupted ERDB or EMDB.	page 93	

Restoring a corrupted ERDB on Redundant Experion Server A

Step	Task	Go to	Done?
1	Use EBR to restore the corrupted ERDB.	page 93	
2	Disable ERDB replication for Experion Server B.	page 95	
3	Restore the ERDB for Redundant Experion Server A.	page 97	
4	Delete the contents of this folder on Experion Server A: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i>	n/a	
5	Use a Station to connect to the primary Experion Server and then synchronize both Experion Servers.	page 100	
6	Re-enter any data that was lost from the time of the previous backup to this restoration.	n/a	

4. Restoring nodes and Experion databases
4.7 Checklists for restoring corrupted Experion Server databases

Restoring a corrupted ERDB on Redundant Experion Server B

Step	Task	Go to	Done?
1	Use EBR to restore the corrupted ERDB.	page 93	
2	Disable ERDB replication for Experion Server B.	page 95	
3	Restore the ERDB for Redundant Experion Server B.	page 97	
4	Delete the contents of this folder on Experion Server B: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i>	n/a	
5	Use a Station to connect to the primary Experion Server and then synchronize both Experion Servers.	page 100	
6	Re-enter any data that was lost from the time of the previous backup to this restoration.	n/a	

Restoring a corrupted EMDb on Redundant Experion Server A

Step	Task	Go to	Done?
1	Use EBR to restore the corrupted EMDb.	Page 93	
2	Disable EMDb replication for Experion Server B.	Page 95	
3	Restore the EMDb for Redundant Experion Server A.	Page 98	
4	Delete the contents of this folder on Experion Server A: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i>	n/a	
5	Use a Station to connect to the primary Experion Server and then synchronize both Experion Servers.	Page 100	
6	Re-enter any data that was lost from the time of the previous backup to this restoration.	n/a	

Restoring a corrupted EMDB on Redundant Experion Server B

Step	Task	Go to	Done?
1	Use EBR to restore the corrupted EMDB.	Page 93	
2	Disable EMDB replication for Experion Server B.	Page 95	
3	Restore the EMDB for Redundant Experion Server B.	Page 99	
4	Delete the contents of this folder on Experion Server B: <i>C:\Documents and Settings\All Users\Application Data\Honeywell\Experion PKS\Checkpoint</i>	n/a	
5	Use a Station to connect to the primary Experion Server and then synchronize both Experion Servers.	page 100	
6	Re-enter any data that was lost from the time of the previous backup to this restoration.	n/a	

4.8 Restoration tasks for the recovery environment

Summary of procedures in this section

Task	Go to
Use EBR to restore a hard disk.	page 85
Use EBR to restore selected files.	page 93
Use the Recovery Point Browser to mount a backup image as a drive.	page 94
Disable ERDB and/or EMDB replication for Redundant Experion Server B.	page 95
Restore the ERDB for Redundant Experion Server A.	page 96
Restore the ERDB for Redundant Experion Server B.	page 97
Restore the EMDB for Redundant Experion Server A.	page 98
Restore the EMDB for Redundant Experion Server B.	page 99
Synchronize the redundant server databases.	page 100
Restore the Events Database for Experion Servers.	page 102

Using EBR to restore a hard disk



ATTENTION

During a restore, the file changes made since the last backup image are lost.

Prerequisites

You need to configure backup for the nodes before restoring the hard disk using EBR.
For more information about backup configuration form, see *Sample backup configuration form* in Chapter 5.5, "Scheduling backups."

To use EBR to restore a hard disk

Step	Action	Go to
1	Restart the computer in the recovery environment.	page 86
2	Set the NIC speed. If the node is connected to a LAN that is configured to allow different baud rates, defaulting to the Auto selection may cause the NIC to use a 10 Mbps transfer rate, even if it supports 100 Mbps or more.	page 87
3	Configure network services.	page 89
4	Restore the hard disk image (includes Restore AnyWhere steps).	page 90
5	Run the post-restore command for non-redundant Experion Servers.	page 92
6	Check for proper system operation for Experion Servers.	page 92

To restart the computer in the recovery environment

Step	Action
1	Insert the EBR Recovery CD into the CD drive, and then restart the computer. The Hit any key to boot from CD... message appears.
2	Press any key to continue. After the computer hardware analysis is completed, usually within 5–7 minutes, the computer restarts in the EBR recovery environment. The End User License Agreement dialog box appears.
3	Click Accept .
4	When the dialog appears asking to start the network services, click No , if you are recovering from removable data. Else, the network services automatically start after 10 seconds.
5	Click the date at the bottom left corner of the Symantec Recovery dialog box, select your appropriate time zone, and then click OK .

To set the NIC speed

Step	Action
1	Click Advanced Recovery Tasks on the Symantec Recovery Disk dialog box.
2	Click Network , and then click Set Network Card Speed . The NicCtrlUI dialog box appears.
3	In the Configure Link Speed/Duplex Mode list, select the appropriate speed for your NIC. If you are using FTE, select 100 Mbs Full Duplex .
4	Click OK to close the dialog box.
5	If an error message appears, network services may not have started. To correct this error, perform the following: a) Wait for network services to start, and then repeat this procedure. b) Return to the Advanced Recovery Tasks dialog box, click Network, and then click Start Network Services. c) Wait for the pointer to change to an hourglass and then back to an arrow, and then repeat this procedure.

To configure a MAC address for new NIC cards

If the node includes a new NIC card you need to update the references to the MAC address that appears on the network card.



ATTENTION

It is very important that you configure network adapter ports on the new NIC card to have the same MAC addresses. Doing so will prevent collisions. In addition, no other system on the network uses this address.

Step	Action
1	From the desktop, right-click My Computer and select Manage .
2	Select Device Manager .
3	Expand Network adapters .
4	Right-click the network adapter and select Properties . The Properties page appears.
5	On the Properties page, select Advanced tab.
6	In the Setting area, click the Locally Administered Address field and enter the 12 digit hexadecimal address in the Value field.
7	Click OK .
8	Close the Network dialog box and restart when prompted.

To configure network services

Step	Action
1	Click Network , and then Select Network Configuration Settings . The Network Adapter Configuration dialog box appears.
2	Click the + button at the end of the IP Address 1 box. The Enter Adapter Address dialog appears.
3	On the Enter Adapter Address dialog box, perform the following: a) For FTE, type the Yellow IP address. b) For standard Ethernet, type the IP address of the NIC. c) Type the subnet mask for the IP address. d) If there is a router between the computer being restored and the network location where the backup image file is stored, type the IP address of the router in the Default Gateway box. e) Click OK .
4	Click to close the Network Adapter Configuration dialog box.
5	a) If your backup images are stored on a server or other network drive, you must map this drive to access the backup images. b) If you are restoring from removable media, stop here and return to Chapter 4.8 , "Restoration tasks for the recovery environment" for more tasks.
6	Click Network , and then select Map Network Drive . The Map Network Drive dialog box appears.
7	Open the Drive list, and then select the drive to be mapped from.
8	In the folder box, type the IP address of the server and the shared folder using this format: \\IPAddress\nameofsharefolder
9	Click Finish .
10	If prompted, type the user name (using the format <i>Domain\UserName</i>) and password used for creating the backup image, and then click OK .

To restore a hard disk using EBR

Step	Action
1	On the Home panel, click Recover My Computer to start the System Restore Wizard. The Welcome to the Recover My Computer Wizard dialog box appears.
2	On the welcome page of the wizard, click Next. The Recover Point to Restore dialog box appears.
3	Select a recovery point that you want to recover. Note: Use the Date, Recovery Point, and Size columns in the table to refine your choice.
4	Click Next. The Target Drive dialog box appears.
5	Select the target drive (the drive to recover). If there is no enough free space to restore the recovery point, press the Shift key, and then select multiple, contiguous destinations that exist on the same hard disk.
6	Click Next. The Restore Anyware dialog box appears.
7	If the hardware for recovery is different, select Use Restore Anyware (for recovery to different hardware). If the hardware for recovery is the same, select the hardware from the options provided.
8	Click Next. The Recovery Options dialog box appears.

4. Restoring nodes and Experion databases

4.8 Restoration tasks for the recovery environment

Step	Action
9	Select the following options. – Verify recovery point before restore. – Check for file system errors. – Restore Master Boot Record (MBR). – Restore original disk signature. – Preserve domain trust token on target drive (recommended). The options that are available depend on the restore destination that you select.
10	Click Next. The Completing Recover My Computer Wizard page appears.
11	Select Reboot when finished option (recommended), and then click Finish.
12	Before the computer restarts (in the recovery environment), remove the EBR Recovery CD from the CD drive.
13	If you have previously performed a Restore AnyWhere, you may be prompted for drivers. If so, browse the I386 folder on the EBR Recovery CD for the drivers.
14	If the new hardware includes a replacement NIC, and you are using FTE, you must re-establish the FTE bindings by removing and re-installing FTE.
	Stop The task is complete. Return to the checklist that led to this task.

4. Restoring nodes and Experion databases

4.8 Restoration tasks for the recovery environment

To run the post-restore command for non-redundant Experion Servers

Step	Action
1	Click Start > All Programs > Accessories > Command Prompt .
2	Type "postrestore", and then press the Enter key. This step copies the backed up server data to the correct folder and starts the services.
3	Click Yes to restart the computer.
	Stop The task is complete. Return to the checklist that led to this task.

To check for proper system operation for Experion Servers

Step	Action
1	After the computer restarts in the recovery environment, log on as an Administrator.
2	Choose Start > Programs > Honeywell > Experion PKS > Server > Start-Stop Experion Server to check if the system is operating as the backup.
	Stop The task is complete. Return to the checklist that led to this task.

To use EBR to restore selected files

Step	Action
1	Choose Start > All Programs > Symantec Backup Exec System Recovery > Recovery Point Browser . The Recovery Point Browser dialog box appears.
2	Go to the location of the backup image to restore from, select the image, and then click Open . The Recovery Point Browser appears showing the backup image.
3	a) Expand the backup image tree node. b) Expand the disk drive or folder that contains the files or folders to restore. c) Click the appropriate files and folders to restore.
4	Click File > Recover Files .
5	Click Browse to search for a destination location for the restored files (if different from the default location), and then click Recover .
6	Click Yes or Yes to All to confirm the restoration.
7	Choose Start > Programs > Honeywell > Experion PKS > Server > Start-Stop Experion Server to confirm that the system is operational.
	Stop The task is complete. Return to the checklist that led to this task.

To use the Recovery Point Browser to mount a backup image as a drive

Step	Action
1	Choose Start > All Programs > Symantec > Backup Exec System Recover > Recovery Point Browser . The Recovery Point Browser and the Open window appear.
2	In the tree pane, click the backup image that you want to mount.
3	Choose File > MountRecovery Point > Drive . The Mount Recovery Point > Drive dialog box appears.
4	Click the appropriate drive letter in the Drive Letter list.
5	Click OK .
	Stop The task is complete. Return to the checklist that led to this task.

Disabling ERDB and/or EMDB replication for Redundant Experion Server B



ATTENTION

This procedure applies only to Experion Server B of a redundant server.

To disable ERDB and/or EMDB replication for Redundant Experion Server B

Step	Action
1	Start Configuration Studio and connect to the system with Experion Server B.
2	In Configuration Explorer , click the server name, and then click Administer the control strategy database . The DBAdmin dialog box appears.
3	Expand the DBAdmin and Server Node trees.
4	- For the ERDB, click ERDB Admin Tasks. - For the EMDB, click Administer the system database > EMDB Admin Tasks.
5	Click Disable Replication , and then click Yes .
6	Click OK to acknowledge the disable replication succeeded message.
7	Close the DBAdmin dialog box.



Stop

The task is complete. Return to the checklist that led to this task.

Restoring the ERDB for Redundant Experion Server A

Use this procedure after restoring a backup of Experion Server A to update it with the latest ERDB from Experion Server B. Besides, you can use this procedure to replace a corrupted ERDB on Experion Server A with a valid ERDB from Experion Server B.

Prerequisite

You must perform the action mentioned in “Using EBR to restore a hard disk” on page [85](#) to restore Experion Server A.

To restore the ERDB for Redundant Experion Server A

Step	Action
1	Start Configuration Studio and connect to the system with Experion Server B.
2	In Configuration Explorer , click the system name, and then click Administer the control strategy database . The DBAdmin dialog box appears.
3	Expand the DBAdmin and Server Node trees, and then click ERDB Admin tasks .
4	Select Recover Secondary Database , and then click Yes .
5	Click OK to acknowledge that the recovery is complete.
6	Click Enable Replication , and then click Yes .
7	Click OK to acknowledge the replication successful message.
8	Close the DBAdmin dialog box.
9	On Experion Server A, synchronize the ERDB with Server B as follows: a) Choose Start > Settings > Control Panel > Administrative Tools > Services . b) Click Experion PKS System Repository , and then click Stop . c) Click Yes to stop the Control Data Access service. d) After the services stop, select Experion PKS Control Data Access Server , and then click Start .
10	Close the Services dialog box.
	Stop The task is complete. Return to the checklist that led to this task.

Restoring the ERDB for Redundant Experion Server B

Use this procedure after restoring a backup of Experion Server B to update it with the latest ERDB from Experion Server A. In addition, you can use this procedure to replace a corrupted ERDB on Experion Server B with a valid ERDB from Experion Server A.

Prerequisite

You must perform the action mentioned in “Using EBR to restore a hard disk” on page [85](#) to restore Experion Server B.

To restore the ERDB for Redundant Experion Server B

Step	Action
1	Start Configuration Studio and connect to the system with Experion Server A.
2	In Configuration Explorer , click the system name, and then click Administer the system database . The DBAdmin dialog box appears.
3	Expand the DBAdmin and Server Node trees, and then click ERDB Admin tasks .
4	Select Recover Primary Database , and then click Yes .
5	Click OK to acknowledge that the recovery is complete.
6	Click Enable Replication , and then click Yes .
7	Click OK to acknowledge the replication successful message.
8	Close the DBAdmin dialog box.
9	On Experion Server B, synchronize the ERDB with Server A as follows: a) Choose Start > Settings > Control Panel > Administrative Tools > Services . b) Select Experion PKS System Repository , and then click Stop . c) Click Yes to stop the Control Data Access service. d) After the services stop, select Experion PKS Control Data Access Server , and then click Start .

4. Restoring nodes and Experion databases

4.8 Restoration tasks for the recovery environment

Step	Action
10	Close the Services dialog box.
	Stop The task is complete. Return to the checklist that led to this task.

To restore the EMDB for Redundant Experion Server A

Step	Action
1	Start Configuration Studio and connect to the system with Experion Server A.
2	In Configuration Explorer , click the system name, and then click Administer the system database . The DBAdmin dialog box appears.
3	Expand the DBAdmin and Server Node trees, and then click EMDB Admin tasks .
4	Click Recover Secondary database , and then click Yes .
5	Click OK to acknowledge the successful restore.
6	Click Enable Replication , and then click Yes .
7	Click OK to acknowledge the successful replication.
8	Close the DBAdmin dialog box.
	Stop The task is complete. Return to the checklist that led to this task.

To restore the EMDB for Redundant Experion Server B

Step	Action
1	Start Configuration Studio and connect to the system with Experion Server B.
2	In Configuration Explorer , click the system name, and then click Administer the system database . The DBAdmin dialog box appears.
3	Expand the DBAdmin and Server Node trees, and then click EMDB Admin tasks .
4	Click Recover Primary database , and then click Yes .
5	Click OK to acknowledge the successful restore.
6	Click Enable Replication , and then click Yes .
7	Click OK to acknowledge the successful replication.
8	Close the DBAdmin dialog box.
	Stop The task is complete. Return to the checklist that led to this task.

Synchronizing the redundant Experion Server databases



ATTENTION

Automatic checkpointing from the primary Experion Server to the backup Experion Server does not occur until the two servers are properly synchronized.

To synchronize the Experion Server databases

Step	Action
1	Start Experion on the primary Experion Server and connect a Station to it.
2	Click View > System Status > Server Redundancy to start the Server Redundancy Status display on the primary Experion Server. In the Link Status group, the LEDs must change from red to green and indicate the following: • Backup server is green. • Link On (LINKOn) is OK.
3	Click Synchronize, and then click Yes. The Experion database on the primary server is automatically copied to the backup Experion Server. The synchronization time depends on the size of the system. After synchronization, the synchronization indicator (first LED in the Backup Server group) changes from red to green and the text changes from Unsynchronized to Synchronized.
4	If you want Experion Server A to be the primary database, click Manual Failover , and then click Yes start the failover.
5	After a few minutes, click Synchronize , and then Yes to resynchronize the databases. (If you start synchronization immediately after the failover, an error might occur.)



Stop

The task is complete. Return to the checklist that led to this task.

4.9 Restoring Experion Server databases from checkpoints

Restoring the Engineering Repository Database and controllers

CAUTION Be sure your process is “Off Control” as these procedures require that the controller be idled and/or a new database loaded.

After restoring the Experion database, the database and controllers may be out-of-sync if the control strategy or controller parameters have changed since the backup was made.

If the control strategy or controller parameters remain the same since the backup, you can update the current parameters by uploading the controller to the ERDB, and then updating the project from loaded. You must also save the checkpoint files again from the controller.

If the control strategy has changed since the backup, or if Qualification and Version Control System (QVCS) is licensed and the QVCS database changed (such as check in and check out), you must re-enter the changes to the ERDB project database after it is restored. Then, the controller must be updated by loading the missing blocks as follows:

1. Add the new blocks to the project side of the ERDB but do not load them yet.
2. For each controller that changed since the backup, load the controller checkpoint that was restored with the database to delete the controller copies of the new blocks. This is necessary because the new blocks are “ghost” points and cannot be deleted directly.
3. Note that if the added blocks were IOMs, the checkpoint load may fail. In this case you must bring the controller to the NODB state by using NTOOLS or pulling the batteries and cycling the controller power, and then loading the checkpoint.
4. Load the new added blocks.
5. If there were external connections to the added blocks, any blocks referencing them in other Control Modules or controllers (if peer-to-peer) must be reloaded to update the references to these blocks.
6. When completed, save the controller checkpoint and back up the updated database.

After restoring Experion Server B of a redundant pair, note the checkpoint files. If older checkpoint files are restored to Experion Server B, synchronization attempts to update them from Experion Server A, causing them to become out-of-sync with the database. The solution is to delete the checkpoint files from Experion Server A after replicating the database and before enabling the ERDB synchronization.

Restoring the events database

Step	Action
1	Explore the <i>C:\Program Files\Honeywell\server\data\evtarch\backup</i> folder to find the appropriate backup file to restore. (This is usually the file with the most recent date and time.) Note the name of this file.
2	Start Microsoft SQL Server Enterprise Manager .
3	Expand the left tree down the following path: <i>Microsoft SQL Servers > SQL Server Group > PC Name > Databases > EMS Events</i>
4	Right-click EMS Events Database , and then click All Tasks > Restore Database .
5	In the Restore Database dialog box that appears, ensure that: • Restore as database is set to EMS Events. • Restore is set to From Device.
6	a) Click Select Devices and add the following new device: C:\Program Files\Honeywell\server\data\evtarch\backup\EMSDailyBackup.dat b) Click OK until you return to the Restore Database dialog box.
7	a) In the Restore Database dialog box, keep these default settings: • Backup number = 1 • Restore backup = set • Database = Complete b) Click the Options tab. c) Select the Force restore over existing database check box. d) Select OK to run the restore.

Step	Action
8	If any errors occur, stop SQL server, and then restart it to ensure that no other computer is connected to the database.
	Stop The task is complete. Return to the checklist that led to this task.

Restoring the EMDB

After restoring the EMDB, it may be out-of-sync with the Experion Server if changes were made to the server, asset, or alarm group definitions after the backup was made. For example, if a server was added or deleted, or an alarm group hierarchy was changed, such changes must be reconstructed.

If there were no changes to the Enterprise Model after the backup, no further steps are necessary.

If there were changes to the Enterprise Model after the backup, the following scenarios are possible:

- **Changes were made but not loaded:** In this case, the restored EMDB and the Experion Servers are synchronized. However, you must reconstruct the changes that were not present on the backup image. When the EMDB is complete, it can then be uploaded to the Experion Servers as usual.
- **Changes were made and loaded:** In this case, the EMDB and the Experion Servers are not synchronized. Consequently, you must reconstruct the changes that were not present on the backup image. After the changes, the EMDB and the Experion Servers are synchronized. However, you must load the Experion Servers to ensure that everything is synchronized.

4. Restoring nodes and Experion databases

4.9 Restoring Experion Server databases from checkpoints

Each element of the Enterprise Model is discrete and must be individually loaded. The following table describes which elements must be loaded based on the type of change made.

Type of change	Element to load		
	<i>System</i>	<i>Assets</i>	<i>Alarm groups</i>
System name	Yes	No	No
Experion Servers (add, delete, rename, properties)	Yes	No	No
Assets (add, delete, rename, parents, properties)	No	Yes	Yes (assets are referenced by alarm groups)
Alarm groups (add, delete, rename, parents, properties)	No	No	Yes

5. Developing the backup plan

5.1 About this section

This section provides suggestions for backing up system data from TPS and Experion nodes to prepare for an unexpected event. It would be beneficial to include system administrators and users in developing the backup plan. You can then incorporate the finished backup/recovery plan into your Operations Policy.

5.2 Backup plan considerations

When developing the backup plan, consider the following:

- What, if any, are the prerequisites before backups are started?
- Which system nodes and data repositories are backed up?
- Are the backups hourly, daily, weekly, or monthly? What time of day?
- What type of backup is required for each node—full or incremental?
- How are the backups tracked when developing the backup plan?
- Where are the backup recovery points stored?
- Are the backups performed correctly?
- How can the backup plan effectively manage the following unexpected events?
 - **Natural events:** Fire, flood, earthquake, storms, and extreme weather conditions.
 - **Technological events:** Hardware, software, power failure; virus, and service denial.
 - **Human events:** Judgment error, theft, vandalism, misconduct, and security breach.

Preventing data loss

One method of preventing data loss because of unexpected events is to perform regularly-scheduled system backups. You must analyze the importance of your data and systems and ensure that an appropriate backup plan is established and performed regularly.

You can monitor system usage and other controls to check the load on system resources such as hard disks and networks. This enables you to perform needed upgrades and preventive scheduled maintenance before a hardware or software failure occurs.

The integrity of your data and systems must be maintained by regularly running virus scans to ensure your systems are not infected.

Prerequisites for a backup

You must know the nodes and node types in your system to determine the following:

- Which nodes are backed up?
- Which drives in each node are backed up?
- Which server is used as the Backup Repository for the backup recovery points?

Backing up the TPS/TPN System and other products

If you are using TPN Backup and Restore to back up images from LCN nodes onto TPS GUS and APP nodes, those images are backed up automatically onto the Backup Repository when the TPS System is backed up by EBR. Ensure that your TPS/TPN backup is done before your scheduled Experion Backup and Restore backup begins.

If there are other software products installed on Experion system nodes, their backup images are included in the EBR backup.

Administrative backup activities

Administrative activities include creating and scheduling backup jobs, monitoring the progress and results of those jobs through logs and event notifications, and controlling the backup jobs. You accomplish these activities using the EBR Manager Console, Desktop User Interface, or Server User Interface.

File naming guidelines

You must be consistent with naming your recovery point files. In addition, the names must be clear and meaningful. This makes the task of restoring data easier. For example, for a node named “ESVR_1,” a good recovery point filename would be “ESRV_1_C” rather than “C_Drive.” This provides for proper identification of the recovery point file if it is copied or exported to another location.

Checkpointing data repositories

When you are scheduling EBR backups, ensure that you co-ordinate your backup scheduling with the manual checkpoints that must be performed for Experion and TPS/TPN repositories. Ensure that manual checkpointing of Experion and TPS/TPN repositories is performed before the Experion Backup and Restore backup jobs begin.

Staggering start times

When scheduling backup jobs for multiple nodes, it is recommended that the start times be staggered so that only one node is being backed up at a time, to minimize loading on the network.

Recovery Point

CAUTION Do not use the hard disk of a Target Node to store backup image files. If the hard disk becomes full, it can impede the node's performance.

Typically, a Target Node does not contain sufficient space to store all the Recovery points. Keeping Recovery points on the Target Node may degrade the performance of the node. In addition, if the hard disk drive in the Target Node fails, the Recovery points are lost.

For security purposes, the server that stores the Recovery points must be located in the same physical area as the network administrators.

The number of nodes in your system determines the number of servers required.

The primary guideline is that each full Recovery points requires about half the amount of space as is being used on the hard disk being backed up.

Another guideline to remember is that less than 1 GB of storage capacity is required for most nodes for a week's worth of incremental backups with infrequent consolidation. PHD servers with large databases may use 1 GB per day.

Offsite storage

The frequency that the backups are taken offsite is determined by the importance of the data and systems. Any mission-critical systems or data (those that must be recovered within minutes or hours of the system failure) must be backed up and taken offsite at least daily. The location you choose is important. The backups must not only be safe and secure, but quickly and easily accessible in the event of a disaster.

You must transfer your backup images onto a DVD or tape (a CD may not contain sufficient capacity) on a regularly-scheduled basis. You must store the media offsite in a secure temperature-controlled environment for long-term storage. You must, in addition, define the backup retention period for the media.

Backup/restore speed

Both the backup and restore procedures transfer data at a rate of 1 Gigabit per minute to a network drive (based on the uncompressed disk space in use). Backup to a USB hard disk mounted locally has similar performance, but recovery time is nearly twice as fast.

For backup and restore to DVD, allow 4 minutes per Gigabit.

For older platforms, the transfer rate may be 0.5 Gigabits per minute to a network drive.

Note that these restore times are the times to restore the hard disk after the 5–6 minutes required to restart the computer from the recovery CD and initiate the task.

The **Verify recovery point after creation** option takes about twice as long to complete.

Honeywell File Transfer jobs require up to 10 percent longer time to complete if performed concurrently with a backup job.

Scanning a hard disk with anti-virus software may require up to 75 percent longer time to complete if performed concurrently with a backup job.

In each of these cases, the impact on the time required for the backup is less than 10 percent.

Note: The approximate data provided is valid in ideal conditions.

Rebuilding or restoring data

Do not rebuild or restore a database at the same time that a backup is being performed. Database corruption can occur when the node is low on hard disk space and you are doing a database rebuild when a backup is in progress.

Running disk I/O intensive applications

Running disk I/O-intensive applications at the same time that a backup is being performed can cause a significant performance impact on the time required to complete the backup and/or to complete the other application. It is best to avoid using these applications at the same time.

Running disk defragmentation software

Do not run disk defragmentation software when a backup is in progress. It causes the backup image to be invalid and you are not allowed to restore the hard drive. In addition, do not run any partitioning software at the same time that a backup is being performed.

Running general applications

Some general applications such as ScanDisk must be disabled when a backup is in progress. You must, in addition, avoid Honeywell-specific actions such as Experion Server synchronization, archiving/restoring to/from a tape, and file transfer to/from the LCN.

Data compression

Using data compression lets enable the recovery points to be saved in a smaller size file.



ATTENTION

The **Standard** option must be used for data compression because it provides the best overall performance results.

EBR provides the following compression options:

- **Standard** compression is recommended because it provides the best combination of backup speed and size. Standard compression results in a backup image file that is about 50 percent (or less) of the size of the space used on the hard disk. For nodes where the majority of hard disk usage is for a PHD database, the backup image file is about 30 percent of the size of the hard disk space in use.
- **Medium** compression increases CPU usage by about 30 percent and results in a backup image file that is about 40 percent of the size of the used space on the hard disk.
- **High** compression keeps CPU usage between 80 and 100 percent, when taking 15 percent longer than standard or medium compression to produce a backup image file that is about 40 percent of the size of the used space on the hard disk. Using higher compression options slightly reduces the recovery time by 5 to 15 percent.
- **None** or no compression takes 60 to 70 percent longer than standard compression (because of the increase in the number of file write operations), but has the advantage of reducing the peak CPU usage by 30 percent.

Using routers

Although you can back up a device through a router, severe performance consequences can occur. Where possible, the backup images must be stored on a device that is on the same LAN segment as the Target Nodes.

5.3 Backup recommendations



ATTENTION

The backup recommendations in this section are only suggestions because Experion and TPS systems differ between organizations. When scheduling full and incremental backups, refer to your organization's Operations Policy.



ATTENTION

The *esv_db_beforecapture.cmd* script invokes the Independent recovery / FULLBKUP utility for Experion Servers. Before using this script in a scheduled backup job, you must determine the time required to complete the Independent recovery / FULLBKUP job.

If the Experion Server contains large event and history databases, an internal full backup of the database may exceed the maximum 15-minute timeout suggested for EBR *beforecapture* scripts. In this case, you must schedule Experion Server Independent recovery / FULLBKUP jobs as distinct jobs that do not overlap the scheduled EBR backup jobs.

Optionally, using the *esv_db_beforecapture.cmd* script is recommended only if Experion Server Independent recovery / FULLBKUP jobs are not scheduled independently of EBR backup jobs. This option leads to a larger incremental image.

Experion Servers

- Full backups run weekly.
- Incremental backups run every 4 hours—consolidate daily.

Experion Stations, Console Extension Stations, Flex Stations, TPS Stations

- Full backups run weekly.

ACE-T, SIM-ACE-T, and SIM-C200 nodes

- Full backups run weekly.
- Incremental backups run every shift change—consolidate daily.

Uniformance applications

Backup scripts are not required for most Uniformance applications because EBR interacts with Microsoft's Virtual Shadow Software (VSS). This ensures that the recovery points on Windows 2003 Server Standard and Windows XP Professional for VSS-compliant databases (including current versions of SQL Server) are consistent without the need for command files.

Process History Database



ATTENTION

- EBR only supports Process History Database (PHD) software installed on a TPS APP Node to collect process history data directly from the LCN. EBR does not support PHD software installed on a separate computer platform.
 - Avoid performing PHD administrative operations during a backup.
-

When using the Configure Jobs Wizard (in the Management Console) to create a backup job for a PHD Server, you can configure the PHD command files to run in one of three modes—before data capture, after data capture, or after creation of the backup image.

To see an example of the [Command Files](#) page of the Configure Jobs Wizard, see page [129](#).

There are two PHD command files:

- The *PHD_BeforeCapture.cmd* file lets two CONSTORE cycles before the script makes CONSTORE inactive. With CONSTORE inactive, the backup is started. Select this file for the **Before data capture** list on the [Command Files](#) page.
- The *PHD_AfterCapture.cmd* file restarts CONSTORE after the PHD backup is complete. Select this file for the **After data capture** list on the [Command Files](#) page.

All command files must be located in the *C:\Documents and Settings\All Users\Application Data\Symantec\Backup Exec System Recovery\CommandFiles* directory on the Target Node. However, the exact path depends on whether the node is a PHD Server or Uniformance Desktop node.

Editing the PHD time-out list

On the [Command Files](#) page, you may need to edit the **Timeout (sec)** list to increase the timeout value for the *PHD_BeforeCapture.cmd* script. (The default value of 60 seconds must be sufficient for execution of the *PHD_AfterCapture.cmd* file.)

The timeout value used with *PHD_BeforeCapture.cmd* depends on these two factors:

- The time taken to archive the PHD configuration database—the delay time used in the *PHDful.cmd* script file (which depends on the CONSTORE cycle time).
- The time taken to copy the PHD archives files to another location. The time required to perform the latter operations may require a timeout of 20 minutes or more.

CAUTION

For the PHD, the delay time plus the time required to copy the archive files to the back location must not exceed the number of minutes defined by PHD parameter DEF_QMINUTE. Depending on the sizes of the data queues, data may be lost if the queues fill up when CONSTORE is disabled.

Typically, data queues are sized for three times the CONSTORE rate; however, you can override this setting.

PHD parameter DEF_QMINUTE determines the data queue sizing in minutes.

PHDMAN SHOW SYSTEM provides you with a way to see the CONSTORE interval. You must ensure that the time that CONSTORE is disabled does not exceed the DEF_QMINUTE value. Otherwise, it is possible that the queues may be overwritten and data lost.

5.4 Types of backups

Drive-based backup

- Back up and recover your computer's system drive (typically, the C drive, which includes your operating system).
- Back up and recover a specific hard drive, such as a secondary drive (which is a drive other than the system drive on which your operating system is installed).
- Recover lost or damaged files or folders from a specific point in time

The drive-based backup enables you perform two types of backups.

1. **Recovery point set** schedules a base recovery point with additional recovery points that contain only incremental changes that were made to your computer since the previous recovery point.
2. **Independent recovery point** creates a complete, independent copy of the drives that you select. This backup type typically requires more storage space, especially if you run the backup multiple times.

5.5 Scheduling backups

Scheduling considerations



ATTENTION

When scheduling backup jobs for multiple nodes, Honeywell recommends staggering the start time therefore only one node is backed up at a time.

Scheduling the frequency of backups normally depends on the following factors:

- Rate of data change (configured data)
- Database availability
- Value of the data to the company

Sample backup configuration form

Table 1 is a sample backup configuration form that you can create as a spreadsheet to simplify backup scheduling by having all the data in one place. After entering the data, you can use the spreadsheet data to configure backups for each node.

Table 1 Sample backup configuration form

Configuration Information	Range/Options/Notes	Node A	Node B
Node Number			
Node Name			
Node Type			
Hardware Platform (optional)			
General backup information:			
• Network file name	Name of backup file		
• Network folder name of Recovery Point			
MAC Address			
IP Addresses:			
• Node (Yellow FTE)			
• Default Gateway			
• File Server			
Scheduling backups (Full or Base with incrementals):	• Once • Weekly • Monthly • No schedule		
• Weekly:			
– Day of the week	Sun, Mon, Tue, Wed, Thu, Fri, Sat		
– Start time	Hours: Minutes AM/PM		
– First backup of each day	Hours: Minutes AM/PM		
– Last backup of each day	Hours: Minutes AM/PM		
• Monthly:			
– Day of the month	1–31		
– Start time	Hours: Minutes AM/PM		

5. Developing the backup plan
5.5 Scheduling backups

Configuration Information	Range/Options/Notes	Node A	Node B
– Number of hours between incrementals	1–?		
Event-driven incrementals	Not recommended.		
Data compression:	• Standard (recommended) • Medium • High • None		
Verify recovery point after creation	Not recommended. Verification doubles the time to finish a backup.		
Divide backup image into smaller files:	Select this option only if you are working with removable media at a later time. Not recommended for backups kept on a file server.		
• File descriptor (for each node backup)			
• File size	• CD: 650 or 700 MB • DVD: 4589 MB • Tape: enter capacity		
Limit number of backups for each hard disk	5 backups maximum per week are recommended. 1 backup at month-end is recommended.		
Consolidate incremental backups	Not recommended unless disk space is limited. • Never • Every 12 hours (recommended) • Every 4 hours • Frequent		

6. Developing the recovery plan

6.1 About this section

This section provides suggestions for restoring system data to TPS and Experion nodes after recovering from an unexpected event. It would be beneficial to include system administrators and users in developing the recovery plan. You can then incorporate the finished backup/recovery plan into your Operations Policy.

6.2 General recovery plan considerations

When developing the recovery plan, consider the following:

- What are the possible unexpected events that can affect your process and system?
- Depending on the unexpected events, what are the possible recovery procedures?
- What is the recovery procedure for each scenario?
- Should you develop a plan for each of the possible recovery scenarios?
- Who performs the recovery tasks?
- If your plant is divided into operational areas, is there a hierarchy of operational areas to be restarted?
- What is the order in which the system nodes within a cluster or operating area are to be restarted?
- Where is your recovery point files stored? Are they available from a nearby secure, temperature-controlled off-site location?

The system recovery plan must be documented and available to all possible participants in the recovery scenarios.

6.3 Elements of a recovery plan

The recovery plan is a comprehensive set of measures and procedures that are implemented to ensure that essential, critical resources and infrastructures are maintained, or that you have backup alternatives in place in case of an unexpected event.

The recovery plan must contain a comprehensive list of actions to be taken before, during and after an unexpected event. Having a recovery plan means that your organization implemented actions that allow you to respond to and recover from an unexpected event in an organized and efficient manner. The recovery plan must be documented, tested, and approved to ensure the actions work as planned.

The recovery plan can protect the organization in a number of ways, including:

- Providing a sense of security.
- Ensuring system and resource stability during an unexpected event.
- Minimizing system downtime and recovery.
- Minimizing the risk of permanent loss of critical systems and data.
- Minimizing confusion during an event.
- Minimizing the amount of decision making after the event.
- Providing a platform on which to test various disaster recovery scenarios.
- Ensuring the reliability of secondary systems such as mirror servers and uninterruptible power supply (UPS) systems.

When developing a recovery plan, you must remember that its instructions may be carried out in extremely stressful conditions. Therefore, it is essential that the plan contains easy-to-follow instructions written in a chronological order.

You must perform the following tasks when you begin developing the recovery plan:

- Analyze the business impact
- Perform a risk assessment
- Develop the recovery plan
- Test and audit the plan

6.4 Overview of tasks to develop a recovery plan

Analyze the business impact

Identify the critical systems, applications and business processes and prioritize them in order of importance. You must include:

- System software
- Experion application software
- Control strategies
- History and event archives
- Displays

You must then evaluate the impact of an interruption on each of the systems, applications and processes you identified as being critical.

Perform a risk assessment

Assessing the risks to your organization help to identify any weaknesses to your critical systems, including the physical security of your building, data security, and the likelihood of a disaster occurring. The risk assessment may help to identify potential risks to your organization and minimize certain threats by identifying system vulnerabilities which you can then address.

Use the recovery plan

Recovery plans helps to minimize downtime of your critical systems. The recovery plans must outline how each of the critical systems, applications, and processes are restored, and provide a list of resources required for plan implementation.

The recovery teams must be prepared to quickly deal with the unexpected event. With this in mind, you must ensure that the recovery plan is concise and clearly written.

After the plan has been written and tested, employees must be notified of its existence and key personnel must be trained in its use. It is important to keep copies of the entire plan in a secure offsite location. The plan must be stored in hard-copy format, as well as electronically in case computer systems are unavailable.

Test and audit the plan

You must audit and test the plan periodically to ensure that it continues to meet the organization's procedures, practices, and policies.

6. Developing the recovery plan

6.4 Overview of tasks to develop a recovery plan

7. Implementing the backup and recovery plan

7.1 About this section

This section provides instructions for using the various EBR Server / Desktop GUIs to create and manage full, incremental, and manual backup jobs.

This section in addition provides information about managing the Backup Repository.



REFERENCE - EXTERNAL

For more information about backup jobs, see the following documentation on the EBR product DVD: Chapter 5, "Managing backups" in the *Symantec Backup Exec™ System Recovery Manager Implementation Guide*.

7.2 Defining a drive-based backup

A drive-based backup takes a snapshot of your entire hard drive, capturing every bit of information that is stored on it for later retrieval. All of your files, folders, desktop settings, programs, and your operating system are captured into a recovery point. You can then use that recovery point to restore individual files or folders or your entire computer.

-
- CAUTION**
- Server synchronization may corrupt a backup. Ensure that the servers do not synchronize during the backup.
 - Backups must be scheduled around the timing of internal checks and procedures performed by the Experion system which may corrupt the backup. To ensure your backup does not conflict with these times, avoid scheduling backups between midnight and 3 AM. These times must be reviewed against the plant operating and administration policies as they could change based on the operating environment and operations specific to your plant.
 - During the backup, ensure that Checkpoint saves are not running in the background or scheduled to run. To check this, choose Control Builder Menus Controller > Checkpoint > Schedule Checkpoint Tasks. Then, click Stop to cancel all running and scheduled tasks.
 - Ensure event archiving is not occurring.
-

Creating a recovery point set using Server/Desktop User Interface

Schedule a base recovery point with additional recovery points that contain only incremental changes that were made to your computer since the previous recovery point.

Incremental recovery points are created faster than the base recovery point. In addition, they use less storage space than an independent recovery point.



ATTENTION

You can only have one recovery point set defined for each drive. The Recovery Point Set option is not available if you have already assigned a selected drive to an existing backup and specified Recovery Point Set as the recovery point type. This option is unavailable if you select an unmounted drive that cannot be part of a recovery point set.

To create a recovery point set, perform the following steps.

Step	Action
1	On the Home page, click Run or Manage Backups .
2	In the Run or Manage Backups dialog box, click Define New . If you have not yet defined a backup, the Easy Setup dialog box appears instead.
3	Click Back up my computer , and then click Next
4	Select one or more drives to back up, and then click Next . Press and hold Ctrl key to select multiple drives. If you do not see an existing drive, check Show Hidden Drives .
5	Perform one of the following: • If you have selected a drive that is already included in a defined backup, click Next, and then skip to step 8. • Click Add drives to an existing backup, select an existing backup in the Select the backup drop-down list, and then click Next. • Click Define a new backup to define a new backup, and then click Next.
6	Select Recovery point set (recommended) to do an incremental backup.
7	Click Next .
8	On the Backup Destination page, select from the following options: • Browse to a location in the Folder field in which you want to store the Recovery points. • Type the username and password in Network Credentials to copy the backup to the network drive. • To customize recovery point file names, click Rename, and type a new file name. • Click Add button to add up to two Offsite Copy destinations.

Step	Action
9	If you want to make copies of your recovery points to store at a remote location for added backup protection, perform the following: • Click Add and then select Enable Offsite Copy. • Select Prompt me to start a copy when I attach an external Offsite Copy destination drive option, if you want recovery points to be automatically copied to external Offsite Copy destination drives whenever you plug one in to your computer. • Click Browse to locate an Offsite Copy destination. • Click Add an additional Offsite Copy destination, if you want to add a second destination, and then specify the path (a local folder, network path, or FTP address) to that destination. • Click OK.
10	Click Next.
	 ATTENTION You cannot use an encrypted folder as your backup destination. You can choose to encrypt your backup data to prevent another user from accessing it.
11	On the Options page, select from the following options. • Type a Name for your backup. • Select a Compression level for the recovery point. • Verify recovery point after creation. • Set the limit for the number of recovery point sets saved for this backup. • Enable search engine support. • Include system and temporary files. • Type a description for Recovery point in the text box. • Select an option in the advanced Option dialog box and then click OK.
12	Click Next.

Step	Action
13	On the Command Files page, select the appropriate items, and then click Next. - For Experion Servers in before data capture, if using the optional <i>esv_db_beforecapture.cmd</i>, set Timeout to at least 900, even in high network loads. - For PHD Nodes in before data capture, if using the optional <i>phd_beforecapture.cmd</i>, set Timeout for to at least 500.
14	Click Next .
15	Choose a Recovery Point set as your recovery point type.
16	Click Schedule if you want the backup to run automatically, according to a schedule. If you want to run the backup only when you start it manually, clear Schedule, and then skip to the next step.
17	Enter a start time and select the days of the week when the backup must run.
18	Click the Start a new recovery point set drop-down list, and then define how frequently a new recovery point set must be started. For example, if you select Monthly, a new base recover point is created the first time the backup runs during each new month.

7. Implementing the backup and recovery plan

7.2 Defining a drive-based backup

Step	Action
19	For advanced scheduling options, such as setting up event triggers that start the backup in response to specific events, click Advanced and configure any of the following options: a) Schedule (Backup time) – Click Schedule, and then select the days and a start time for the backup to run. – Check Run more than once per day, if you frequently modify data that you want to protect. – In addition, specify the maximum time between backups and the number of times per day that the backup must run. – In the Automatically optimize drop-down list, select the option to define how often optimization must occur to help manage the disk space that is used by your backup destination. – In the Start a new recovery point set drop-down list, select the option to define how frequently a new recovery point set must be started. – Click Custom to customize the option you select. b) Event Triggers (General) Select the type of events that must automatically start the backup.
20	Click OK, and then click Next.
21	If you want to run the new backup immediately, click Run Backup Now. This option is not available if you configured an independent recovery point with the option to run it only once.
22	Click Finish.

Creating an independent recovery point using Server/Desktop User Interface

Independent Recovery point creates a complete, independent copy of the drives that you select. This backup type typically requires more storage space, especially if you run the backup multiple times.

Step	Action
1	On the Home page, click Run or Manage Backups .
2	In the Run or Manage Backups dialog box, click Define New . If you have not yet defined a backup, the Easy Setup dialog box appears.
3	Click Back up my computer , and then click Next .
4	Select one or more drives to back up, and then click Next . Press and hold Ctrl key to select multiple drives. If you do not see an existing drive, check Show Hidden Drives .
5	Perform one of the following: • If you have selected a drive that is already included in a defined backup, click Next, and then skip to step 8. • Click Add drives to an existing backup, select an existing backup from the Select the backup drop-down list, and then click Next. • Click Define a new backup to define a new backup, and then click Next.
6	Select Independent recovery point to create complete independent copy of the drives that you select.
7	Click Next .
8	On the Backup Destination page, select from the following options: • Browse to a location in the Folder field in which you want to store the Recovery points. • Type the username and password in Network Credentials to copy the backup to the network drive. • To customize recovery point file names, click Rename and type a new file name. • Click Add button to add up to two Offsite Copy destinations.

Step	Action
9	If you want to make copies of your recovery points to store at a remote location for added backup protection, perform the following: • Click Add, and then check Enable Offsite Copy. • Select the Prompt me to start a copy when I attach an external Offsite Copy destination drive option, if you want recovery points to be automatically copied to external Offsite Copy destination drives whenever you plug one in to your computer. • Click Browse to locate an Offsite Copy destination. • Click Add an additional Offsite Copy destination, if you want to add a second destination, and then specify the path (a local folder, network path, or FTP address) to that destination. • Click OK.
10	Click Next.
	 ATTENTION You cannot use an encrypted folder as your backup destination. You can choose to encrypt your backup data to prevent another user from accessing it.
11	On the Options page, select from the following options. • Type a Name for your backup. • Select a Compression level for the recovery point. • Verify recovery point after creation. • Select the Limit the number of recovery point sets saved for this backup. • Enable search engine support. • Include system and temporary files. • Type a description for Recovery point in the text box. • Select an option in the Advanced Option dialog box and then click OK.
12	Click Next.

7. Implementing the backup and recovery plan

7.2 Defining a drive-based backup

Step	Action
13	On the Command Files page, select the appropriate items, and then click Next. - For Experion Servers in before data capture, if using the optional <i>esv_db_beforecapture.cmd</i>, set Timeout to at least 900, even in high network loads. - For PHD Nodes in before data capture, if using the optional <i>phd_beforecapture.cmd</i>, set Timeout for to at least 500.
14	Click Next.
15	Choose an Independent Recovery point as your recovery point type. In the Automatically create a recovery point drop-down list, select one of the following options: No Schedule option - runs the backup only when run it manually. Weekly option - runs the backup at the time and days that you specify. Monthly option - runs the backup at the time and days that you specify. Only run once option - runs a backup one time on the date and at the time you specify.
16	If you want the backup to run automatically according to a schedule, click Schedule. If you want to run the backup manually, clear Schedule and skip to the next step.
17	Enter a start time and select the days of the week when the backup must run.
18	Click the Start a new recovery point set drop-down list, and then select how frequently a new recovery point set must be started. For example, if you select Monthly, a new base recovery point is created the first time the backup runs during each new month.

7. Implementing the backup and recovery plan

7.2 Defining a drive-based backup

Step	Action
19	For advanced scheduling options, such as setting up event triggers that start the backup in response to specific events, click Advanced and configure any of the following options: • Schedule (Backup time) – Click Schedule, and then select the days and a start time for the backup to run. – Check Run more than once per day if you frequently modify data that you want to protect. – In addition, specify the maximum time that must occur between backups and the number of times per day that the backup must run. – In the Automatically optimize drop-down list, select how often optimization must occur to help manage the disk space that is used by your backup destination. – In the Start a new recovery point set drop-down list, define how frequently a new recovery point set must be started. Click Custom to customize the option you select. • Event Triggers (General) Select the type of events that must automatically start the backup.
20	Click OK , and then click Next .
22	Click Finish .

7.3 Running a backup job manually

- CAUTION**
- Server synchronization may corrupt a backup. Ensure that the servers do not synchronize during the backup.
 - Backups must be scheduled around the timing of internal checks and procedures performed by the Experion system which may corrupt the backup. To ensure your backup does not conflict with these times, avoid scheduling backups between midnight and 3 AM. These times must be reviewed against the plant operating and administration policies as they could change based on the operating environment and operations specific to your plant.
 - During the backup, ensure that Checkpoint saves are not running in the background or scheduled to run. To check this, choose **Control Builder Menus Controller > Checkpoint > Schedule Checkpoint Tasks**. Then, click **Stop** to cancel all running and scheduled tasks.
 - Ensure event archiving is not occurring.
-

To run and schedule a backup job using Server/Desktop User Interface

Step	Action
1	On the home page, click Run or Manage Backups . The Run or Manage Backups dialog box appears.
2	Select the backup jobs, and then click Run now . The Progress and Performance dialog box appears.
3	Set the Performance slider to midpoint for optimal performance, and then click Close when the backup is complete.

To run and schedule a backup job manually using Manager Console

Step	Action
1	On the Manage page , click Backup Job , under Create . The Define BackupJob Wizard appears.
2	Select any of the following Recover Points: • Recovery Point Recovery point set (recommended) • Independent recovery point
3	Click Next .
4	On the Backup Destination page, select or create a new backup destination.
5	On the Options page, select the recovery point options, and then click Next .
6	On the Command Files page, select the appropriate items, and then click Next . • For Experion Servers, if you are using the optional <i>esv_db_beforecapture.cmd</i>, set Timeout to at least 900, even in high network loads. • For PHD Nodes, set Timeout for <i>phd_beforecapture.cmd</i> to at least 500. Note: If the command files have been copied to the <i>Symantec\Backup Exec System Recovery\CommandFiles</i>, prior to deployment of the agent, they are automatically copied to each Target Node as part of the deployment step.
7	Click Next .
8	Select Schedule if you want the backup to run automatically, according to a schedule. If you want to run the backup only when you start it manually, clear Schedule , and then skip to the next step. Note: If you chose an independent recovery point as your recovery point type, from the Automatically create a recovery point drop-down list, select one of the given options.
9	Click Next . The Completing Define Backup Job Wizard appears.

Step	Action
10	Click Finish .
11	If you want to run the new backup immediately, click Run backup now , and then click Finish .

7.4 Disabling or deleting backup jobs



ATTENTION

If backing up the hard disk to DVD, you may be prompted to mount a blank DVD. Clicking **Cancel** on this dialog box only cancels the mounting of the DVD but *not* the backup job. If you cancel the backup job at this point without canceling the mounting of the DVD, the agent holds the DVD drive open for the backup job, thus disabling it for other tasks.

To disable a backup job using Server/Desktop User Interface

Step	Action
1	On the home page, click Run or Manage Backups .
2	Select the backup that you want to turn off.
3	Click Tasks > Disable Backup .

Repeat this procedure to re-enable the backup. The Enable Backup menu item changes the selected backup (disabled) to Enable Backup.

To delete a backup job using Server/Desktop User Interface

Step	Action
1	On the home page, click Run or Manage Backups .
2	Select one or more backups, and then click Remove .
3	Click Yes .

7.5 Managing the Backup Repository



REFERENCE - EXTERNAL

For more information about managing the Backup Repository (using the Recovery Point Browser and verifying backup images), see the related topics in the following documentation included on the EBR product DVD:

- [Symantec Backup Exec™ System Recovery Manager Implementation Guide](#)
 - [Symantec Backup Exec™ System Recovery](#)
-

Using the Recovery Point Browser

The Recovery Point Browser enables you to:

- Audit recovery point.
- Verify recovery point.
- Delete recovery point.
- Archive/export recovery point.
- Restore archived recovery point.

For examples of how to use the Recovery Point Browser, see the following topics:

- “To use EBR to restore selected files,” page [93](#).
- “To use the Recovery Point Browser to mount a backup image as a drive,” page [94](#).

Verifying recovery points

Whether or not a backup image was created with the **Verify recovery point after creation** option selected, you can determine whether a backup image file or set of files is valid or corrupt using the Recovery Point Browser. Verification at this level consists of checksums and other integrity checks. It is less thorough and does not require as much time as the verification that is performed during the backup.

Deleting recovery points

You can delete backup images when they are no longer needed or when you want to free-up space in the backup location. Normally, the number of backup images maintained is determined by your backup plan. Manual deletion must be limited to special cases.

CAUTION Normally, the Backup History tab in the advanced view of the Recovery Manager Console reflects the history of backup image creation. Backup images are files that the users can delete with proper permission using Windows Explorer.

Note that the backup history is not updated to indicate that a user deleted the backup image. If any file that is part of a backup set (a base backup image with a set of incrementals) is deleted, data from the remaining backup files in the set is not recoverable.

To delete recovery points

Step	Action
1	Use Windows Explorer to go to the .v2i (full or base backup) or .iv2i (incremental backup) file that you want to delete.
2	Select the backup image file to be deleted.
3	Press the Delete key.
	The backup image file is deleted from the list.

Archiving backups to tape

EBR does not support saving backup image files directly to a tape drive. To do this, you must first save the backup image to a hard disk, and then use your existing backup plan to save the backups from the hard disk to tape.

7. Implementing the backup and recovery plan

7.5 Managing the Backup Repository

8. Multiple Station load based on common image

8.1 Overview

With TPS R4XX and Experion PKS R3XX, you can load multiple Windows based TPS and Experion nodes using a common image. This helps you to reduce the time taken for initialization, system software installation and user configuration repeatedly on each node. The Node Cloning solution package enables you to replicate a node setting on multiple nodes using the backup image on Honeywell Custom Recovery CD.

You can perform the multiple station load using

- Network-based infrastructure
- Network-less infrastructure

This functionality is applicable for the following TPS and Experion nodes.

- Console stations
- Flex
- GUS
- APP
- ES-T
- ACE
- ACE-T



ATTENTION

The procedure and the scripts for Node Cloning are only to be used while node cloning. This procedure must not be used while using Backup and Restore functionality to restore to a single destination node.

Prerequisites

The following infrastructure is required for cloning the image.

- Master node
- Storage device
 - DVD and USB drive for network-less infrastructure.
 - Network connectivity, Domain Controller/File Server/Centralized node for storing the image for network-based infrastructure.
- New multiple nodes ready for cloning.
- Honeywell Custom Recovery CDs for parallel implementation. You must have a CD for each cloning node.
- EBR license information must be available for the EBR installation on master node.
- Node Cloning Checklist (For the checklist see “Sample Node Cloning checklist.”)

8. Multiple Station load based on common image

8.1 Overview

Sample Node Cloning checklist

	General						Licenses		Conditionals		Experion Application Media		MAC Addresses	
	Master Node Image	Storage Device	EBR Recovery CD	Server Name (Experion nodes)	NIC Link Speed	Static IP Address (if any)	EBR License	Windows Product Key	Network Conditionals (optional)	Domain Conditionals	License No.	Authori-zation No.	Yellow	Green
Node name														
Example														
GUS 24	GUS 01	✓	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓	✗	✗
EST 17	EST 03	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

8.2 Setting up a Master node for cloning

Checklist for setting up the Master node

The following tasks are to be performed on the Master Node (The Node from which the image is to be taken).

Step	Task	Go to	Done?
1	Set up a Master node	Page 145	
2	Install Experion Backup and Restore (EBR) Software Package	Page 145	
3	Prepare the Master node for Cloning	Page 146	
4	Take a back u a One Time Backup	Page 146	
5	Copying the disk image onto a storage media	Page 149	
<i>Deploying the cloned image on the destination node</i>			
6	Start the destination computer by using the recovery Environment	Page 150	
	Deploying Master Image	Page 150	
<i>Configure user-specific and computer-specific settings</i>			
7	Post Deployment steps for Experion nodes	Page 156	
	Post Deployment steps for TPS nodes	Page 170	

8. Multiple Station load based on common image

8.2 Setting up a Master node for cloning

Setting up the Master node



ATTENTION

Perform all the installations on C drive only. This is necessary as only the C drive image is deployed on the multiple destination computers. Any installations or any data present on any other drive other than the C drive will not be present on the destination computer.

Perform the following to set up a Master node

Step	Action
1	On a clean machine perform the following installations: a) Install Windows XP or Windows Server 2003 Operating system using ExpPlus DVD b) Install Honeywell TPS / EPKS software applications c) Install Node definition tool for EPKS
	REFERENCE - INTERNAL For information about installing the Node definition tool, see "Installing the Node Definition Tool" section of the <i>Node Definition Tool User's Guide</i>.
	d) Install additional software (as required)
2	Perform the required configuration to deploy the image on multiple destination computers.

Installing Experion Backup and Restore software Package

EBR contains Server and Desktop software packages designed for installation on nodes with Windows operating systems.

For EBR Server software package installation, see Chapter [2.6](#), "Installing the EBR Server software package."

For Desktop software package installation, see Chapter [2.7](#), "Installing the EBR Desktop software package."

Preparing the Master node for Cloning

Perform the following steps to prepare master node for cloning.

Step	Action
1	Run the following command on the command prompt: C:\HWIAC\backup\ > prompt
2	Type <i>Before_Backup_For_Cloning</i> and press Enter . The following message appears: Pre-cloning operations completed successfully.
3	Click OK .

Taking a backup of the master node

Perform the following steps to run a one time backup.

Step	Action
1	Start Symantec Backup Exec System Recovery application. Choose All Programs > Symantec > Backup Exec System Recovery > Backup Exec System Recovery
2	Click Task menu. The Task page appears.
3	Click One Time Backup . The Welcome to the One Time Backup Wizard dialog box appears
4	Click Next . The Drives page appears.
5	Select C:\ drive where OS and other Honeywell Software are installed to back up. Click Next . The Backup Destination page appears.

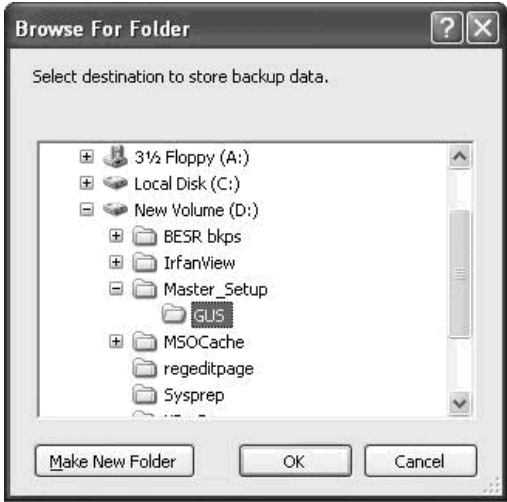
8. Multiple Station load based on common image

8.2 Setting up a Master node for cloning

Step	Action														
6	On the Backup Destination page, select from the following options: • Folder field: Browse to the location in which you want to store the recovery points. You will receive alerts, if Backup Exec System Recovery detects that this location does not contain enough space. You must choose another location that has more space. Click the Browse button. The Browse For Folder dialog box appears. – For network-based infrastructure browse to a shared folder in the domain controller or a file server. – For network-less infrastructure browse to the D:\ drive on the same computer or to any other removable storage devices such as USB drive or USB stick.														
7	Create a new folder by name <i>Master_Setup</i> and share the folder. Inside this folder create a subfolder with the following naming convention: <table><tr><th>Master setup node</th><th>Subfolder name</th></tr><tr><td>Console station</td><td>Console</td></tr><tr><td>Flex station</td><td>Flex</td></tr><tr><td>GUS</td><td>GUS</td></tr><tr><td>APP</td><td>APP</td></tr><tr><td>ACE</td><td>ACE</td></tr><tr><td>ACE-T</td><td>ACET</td></tr></table>	Master setup node	Subfolder name	Console station	Console	Flex station	Flex	GUS	GUS	APP	APP	ACE	ACE	ACE-T	ACET
Master setup node	Subfolder name														
Console station	Console														
Flex station	Flex														
GUS	GUS														
APP	APP														
ACE	ACE														
ACE-T	ACET														

8. Multiple Station load based on common image

8.2 Setting up a Master node for cloning

Step	Action
	
	Network Credentials: In network-based infrastructure, store the recovery point on a network share. In this case, type the user name and password for network access.
8	Click Next. The Options page appears.
9	On the Options page, select from the following options: • Compression: Click Standard in the list. • Verify recovery point after creation: Select this option to automatically test whether a recovery point or set of files is valid or corrupt. • Description: Type a description for the recovery point. The description must help you to further identify the recovery point's contents. For example, you can type in the following information: – Windows XP or Windows Server 2003 (Operating System installed) – Service packs installed – Drivers installed – Honeywell software applications (TPS/EPKS) installed – Type of the node (GUS/APP/Flex/Console/ES-T/ESVT/ACE/ACE-T)

8. Multiple Station load based on common image

8.2 Setting up a Master node for cloning

Step	Action
10	Click Next . The Command Files page appears.
11	Click Next . Note: Do not update any fields in the Command Files page as none of the command files are used for creating Master node backup. The Completing One Time Backup Wizard page appears.
12	Click Finish to run the backup. The Progress and Performance dialog box appears. Move the slider close to Fast (recommended).
13	After the backup operation is completed the status appears as Succeeded . Click Close .

Copying the disk image onto a storage media

Network-based infrastructure: Copy the folder *Master_Setup\node* containing the disk image to a shared folder in a centralized node or a File server or a Domain Controller which resides on the same network as that of the multiple destination computers ready for installation.

Network-less infrastructure: Copy the folder *Master_Setup\node* containing the disk image to removable storage media such as DVD, USB drive and so on for restoring the image to multiple destination computers ready for installation.

8.3 Deploying the cloned image on the destination nodes

Starting the destination computer by using the recovery Environment

For starting the destination computer using the EBR Recovery CD, see the section “To restart the computer in the recovery environment.”



ATTENTION

If the backup image is on a removable media, attach the device to the computer before starting it for the recovery environment to detect it.

Deploying the Master Image



ATTENTION

- Any change made to the Master node after taking the disk image is not reflected in the destination node.
 - Ensure that the destination computer's primary partition capacity is more than the primary partition capacity of the Master node.
-

Perform the following steps to deploy the Master image.

Step	Action
1	On the Home panel, click Recover My Computer to start the System Restore Wizard . The Welcome to the Recover My Computer Wizard dialog box appears.
2	Click Next . The Recovery Point to Restore dialog box appears.

8. Multiple Station load based on common image

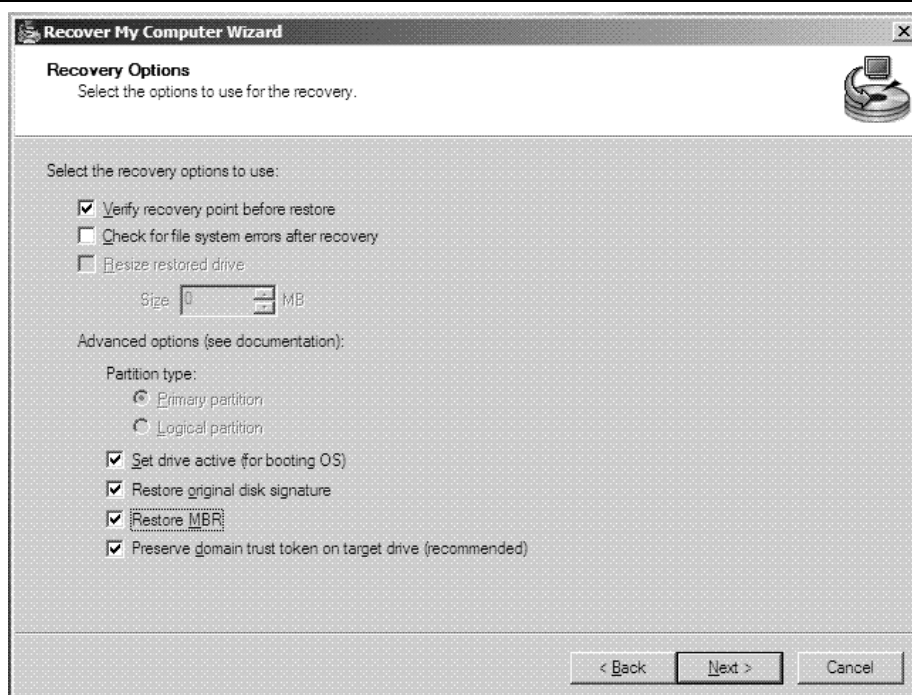
8.3 Deploying the cloned image on the destination nodes

Step	Action
3	From the View by list, click Filename. - Click Browse, locate and select a recovery point (.v2i file), and then click Open. - For network-based deployment, select a network location and type your network credentials. - For network-less deployment, browse to the removable storage media. - Click Next. The Target Drive dialog box appears.  ATTENTION - EBR uses the maximum available network bandwidth for network-based deployment. - You must identify the correct drive based on its label or by the name assigned to it. As in the recovery environment, the drive letters and labels might not match with what appears in the Windows.
4	Select the target drive to restore the disk image. On most of the machines, the default drive is the C drive.
5	Click Next. The Restore Anyware dialog box appears.
6	Select Use Restore Anyware (for recovery to same/different hardware) option (Recommended).  ATTENTION Selecting the Use Restore Anyware option is mandatory for restoring the image of the Master node to bring up a new node. This option is also mandatory when restoring the image on a similar/different platform.
7	Click Next. The Recovery Options dialog box appears.

8. Multiple Station load based on common image

8.3 Deploying the cloned image on the destination nodes

Step	Action
------	--------



- 8 Select the following options:
- Verify recovery point before Restore
 - Restore original disk signature
 - Set drive active (for booting OS)
 - Restore Master Boot Record (MBR)
 - Preserve domain trust token on destination

Note: The options that are available depend on the restore destination you have selected.

- 9 Click **Next** to review the selected restore options.

The **Completing the Recover My Computer Wizard** dialog box appears.

8. Multiple Station load based on common image

8.3 Deploying the cloned image on the destination nodes

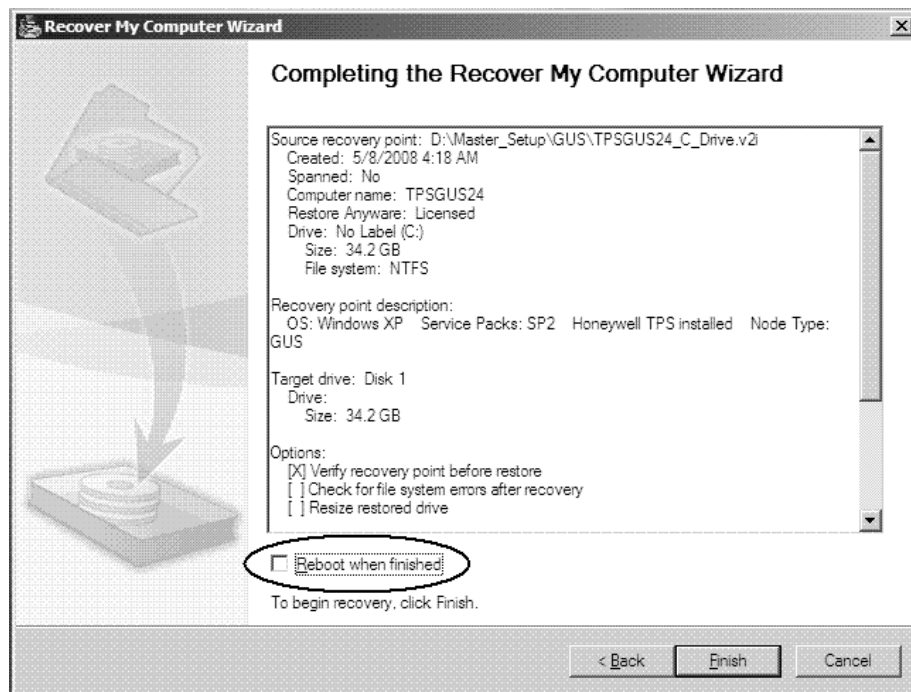
Step

Action



ATTENTION

Ensure that Reboot when finished option is not selected, as a script After_Restore_For_Cloning.vbs has to be run after the image restoration is complete and before the node restarts.



- 10 Click **Finish**.

The **Recover My Computer Wizard** confirmation dialog box appears.

- 11 Click **Yes**.

The **Recovery My Computer Wizard – Progress** dialog box appears.

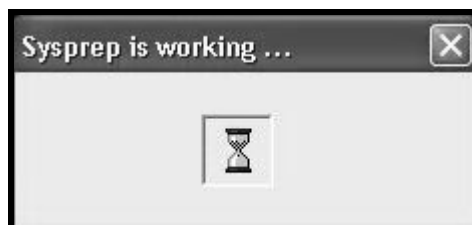
- 12 Click **Close** when the recovery status is complete.

Click **Cancel** if you want to stop the recovery operation.

8. Multiple Station load based on common image

8.3 Deploying the cloned image on the destination nodes

Step	Action
13	On the left pane, click Analyze and then on the right pane click Open Command Shell Window. The command prompt appears.
14	Type C:\HWIAC\backup\Before_Reboot_For_Cloning and press Return / Enter key to run the script. The Before Reboot dialog box appears.
15	If you have a Microsoft Windows XP installation, select Yes. If you have a Microsoft Windows Server 2003 installation, select No. The Post-cloning operations completed successfully message appears.
16	Click OK. Type Exit on the command prompt and return to the home page.
17	On the Home page, click Exit and click Yes on the confirmation message to Exit. The node restarts automatically. The Windows mini setup appears. Then the following dialog boxes appear: – Installing Devices dialog box – Network Settings dialog box – Performing final tasks dialog box Again the node restarts automatically.
18	Log on to the computer as the administrator. The After_Restore_For_Cloning utility starts running. The Sysprep is working... – busy message appears.



The Sysprep takes a few minutes to complete its tasks. Then the node restarts automatically and the Windows mini setup appears.

8. Multiple Station load based on common image

8.3 Deploying the cloned image on the destination nodes

Step	Action
19	When the Microsoft End User License Agreement dialog box appears, ensure to select I accept this agreement option. Click Next. Note: It is mandatory to accept the license agreement as it is a legal aspect of cloning. The Personalize Your Software dialog box appears.
29	Enter the name and organization details and click Next. The Your Product Key dialog box appears.
21	Enter product key for windows and click Next button. The Computer Name and Administrator Password dialog box appears.
22	Enter the new computer name and select Next. The Network Settings dialog box appears. Then the Workgroup or Computer Domain dialog box appears.
23	Select Yes, make this computer a member of the following domain to join the domain. Then enter the domain name and click Next. The Join to Domain dialog box appears.
24	Logon as administrator. Note: You must have administrative privileges to log on as an administrator. The Performing final tasks dialog box appears and the machine restarts automatically.
25	The Registry Cleaning Utility - Regclean starts running. When the message Registry Clean operations completed successfully appears. Click OK.
	Stop The task is complete. Return to the checklist that led to this task.

8.4 Post deployment steps for Experion nodes

Checklist for creating new node definitions and commissioning the destination nodes

Task	Go to	Done?
Install the Node Definition Tool (offline mode)	Page 156	
Create new node definition	Page 157	
Configure node information using the EPC Tool	Page 159	
Configure station connections for Flex Station (if applicable)	Page 165	
Create an options disk	Page 166	
Run the commissioning tool	Page 167	

Launching the Node Definition Tool

Launch the Node Definition Tool (offline mode)

Step	Action
1	Choose Start > Programs > Honeywell Experion PKS .
2	Click System Management and then Offline Node Definition Tool .
Note: Offline is indicated in the Title bar.	

Launch the Node Definition Tool (online mode)

Step	Action
1	Open Configuration Studio and connect to an Experion Server.
2	Expand the network tree at the system level to display computers.
3	Select a computer.
4	Select Launch Node Definition Tool from the right task pane

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Creating new node definition

Step	Action
1	Click Create New Node Definition. Note: In online mode, if the button is unavailable, it indicates a node definition for the selected computer already exists. You must delete the existing node definition from the database before creating a new one.
2	If you are using the tool in offline mode, enter a computer name - this is the default name for the node definition. Note: In online mode, the name of the node definition is the name of the computer selected from the network tree in the Configuration Studio.
3	Select a specific type of node from the Node Type drop down list. Note: This selection determines the software and configuration options available in the subsequent dialog boxes.
4	Select Clone from Computer option as a source for the node configuration information. Other options are as follows: • Use Default Values: Use when a definition for the node type does not yet exist. • Clone from Current Node Definition: Use this option when in offline mode after configuring and saving a node definition of the same node type in the working folder. This is useful when you are configuring several node definitions for the same node type. • Clone from EMDb: Use in online mode only after you have configured and saved a node definition of the same node type to the EMDb. • Clone from Offline Folder: Use when you have configured a node definition for a computer of the same node type. • Clone from Computer: Use when a computer of the same node type is available. Click Browse and navigate to a computer within your network.

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Step	Action
5	Click OK .
	TIP The Configuration Tasks list is populated with tasks relevant to the node type you are defining. Each task is automatically launched in order, allowing you to define specific node options and save the configuration information. You will return to the main dialog box, when you complete all the tasks.
6	If you have selected Flex Station as your node type, go to the "Configuring the station connections for a Flex Station."

Configuring Optional Software for Experion Station – TPS (ES-T) nodes

Step	Action
1	Insert the Experion Application Media. The optional software window appears. Enter the following information in the Install Time License fields: • Name • Company • License No. • Authorization No.
2	To select the Experion App media Path browse to select a path and click Validate Experion App Media Path and License Information . A confirmation dialog box appears.
3	Click OK .
4	Under Select Optional Software , select all the packages that are already installed and the packages that need to be installed. Click Save . Note: If a package is already installed and is not selected it will be removed. The selections have been saved message appears. Click OK .
5	On the Optional Software dialog box, click Exit .

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Configuring node information using the EPC Tool

Use this procedure to configure all the options from the various tabs in the EPC tool. Many of the options are pre-selected based on the source of your node definition.

Step	Action
1	From the Experion PKS Configuration Tool click the Platform tab and select the appropriate options for your node.

Experion PKS Configuration Tool /ANCIM

Platform | Network Configuration | Input/Output Devices | System Details | Activation | Summary

PROCESS BUSINESS
ASSETS PEOPLE

☐ Options Disk Generated for this machine

Installation

☐ Operating System Only

☒ Operating System and Experion PKS (R300/R301)

Media Information

Version: 301.3

Part Number: 51153964

Operating System

☒ Windows XP Professional

☐ Windows Server 2003

Client Platforms

☐ Dell Precision 340

☐ Dell Precision 360

☐ Dell Precision 370

☐ Dell Precision 470

☒ Dell Precision 490

☐ Integrated Operator Node (I.O.N.)

Service Tag

Service Tag: HJT2R1S

Exit About Help



TIPS

Type `ipconfig /all` at the DOS prompt to determine the MAC address of the destination node after restoring the disk image.

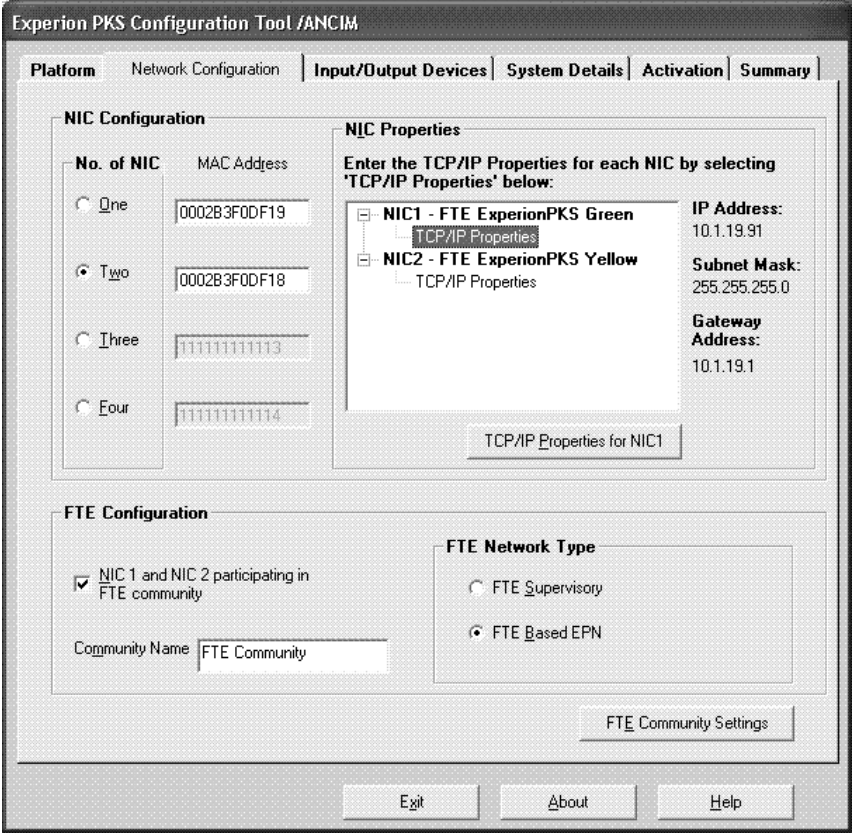
8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Step	Action
2	Click the Network Configuration tab. • Go to step 3 if the node type you are defining is an FTE node. • Otherwise, go to step 4.
3	Define configuration for FTE nodes: • For the NIC configuration, select Two. • Type in the MAC address at this time or use default values and read the values in at a later time (do not include dashes in the MAC address). • Check NIC1 and NIC2 participating in FTE Community (this pre-configures some of the TCP/IP Properties required for FTE). • Type in an FTE Community Name. • Select one of the TCP/IP Properties and then click the TCP/IP Properties for NIC. • Configure the TCP/IP properties. • Click OK when finished. • Click FTE Community Settings.
4	Define configuration for non-FTE nodes: • Select the number of NICs available. • Type in the MAC address at this time or use default values and read the values at a later time. • Click TCP/IP Properties and then click the TCP/IP Properties for NIC. • Configure the TCP/IP properties. • Repeat for second NIC. • Ensure that correct renaming of Green and Yellow NICs is done. • Click OK when finished. • Click System Management Settings. • Select the appropriate FTE Network Type.

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

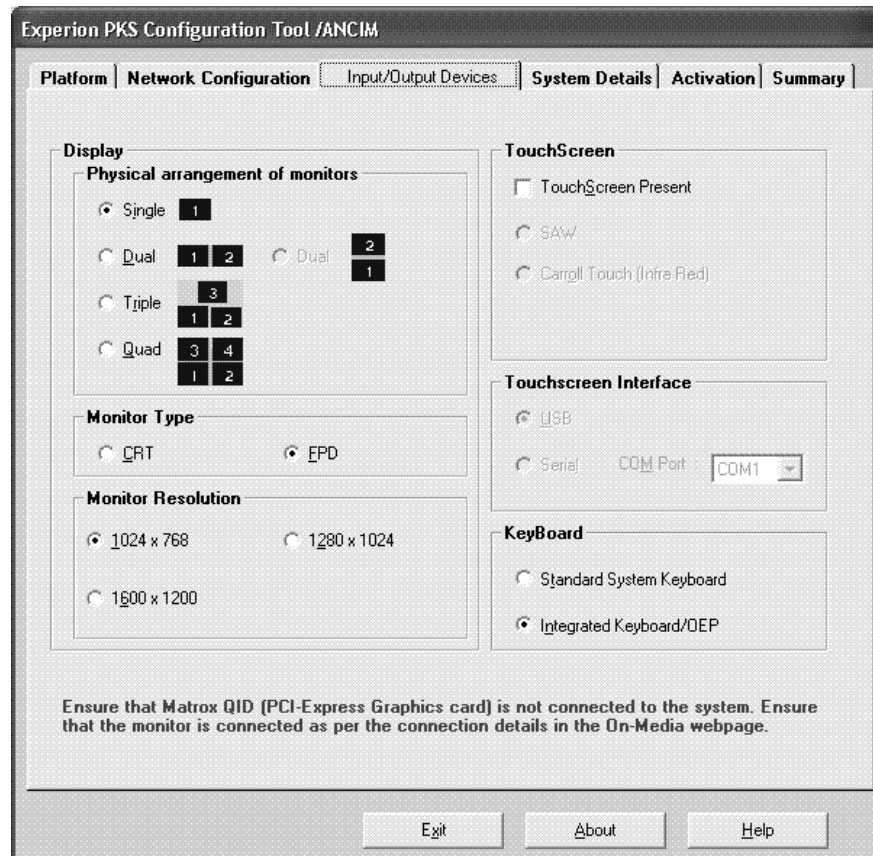
Step	Action
	 The screenshot shows the 'Experion PKS Configuration Tool /ANCIM' window with the 'Network Configuration' tab selected. The 'NIC Configuration' section has 'Two' selected for the number of NICs, with MAC addresses 0002B3F0DF19 and 0002B3F0DF18. The 'NIC Properties' section shows 'NIC1 - FTE ExperionPKS Green' with IP Address 10.1.19.91 and 'NIC2 - FTE ExperionPKS Yellow' with Subnet Mask 255.255.255.0 and Gateway Address 10.1.19.1. The 'FTE Configuration' section shows 'NIC 1 and NIC 2 participating in FTE community' checked, with Community Name 'FTE Community' and 'FTE Network Type' set to 'FTE Based EPN'.
5	Click the Input/Output Devices tab and select the appropriate options for your node.

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Step

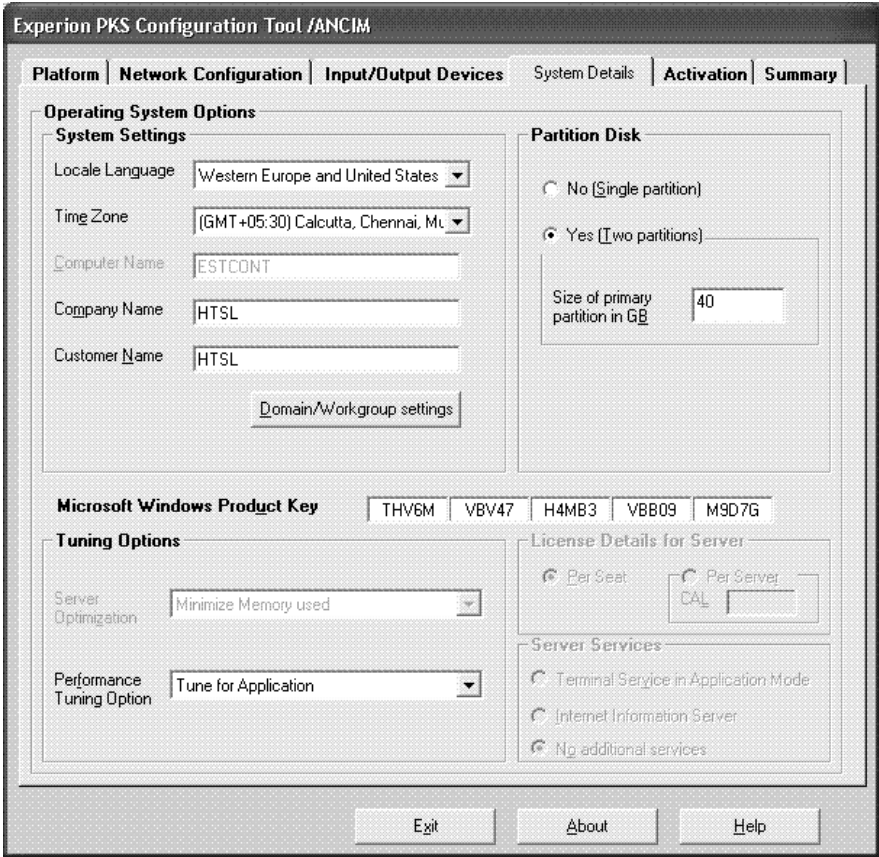
Action



- 6 Click the **System Details** tab.
- 7 From the **System Details** tab, select the appropriate options for your Node Settings.

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Step	Action
	
8	If you are creating an option disk and want to automate the installation process: • Click Domain/Workgroup settings. • Enter the Domain name • Type in a domain administrator name and password that has permission to add a computer to the domain. Note: If you do not enter this information at this time, you will be prompted for it when you perform an installation.
9	Type in the Windows Product Key.

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Step	Action
10	On the Experion PKS configuration Tool/ANCIM dialog box, click the System Details tab, complete the rest of the option selections appropriate for your node type.
11	Click the Summary tab, and then click Process Software . The Experion System Info dialog box appears.
12	Under Set System Type , click the appropriate node type. If the node is an ES-T node, the Experion PKS – Network Selection dialog box appears.
Procedure for an ES-T node	
13	Under Server/Client Network , select FTE and under Optional Control Network connections select LCN . Click Next . The Experion PKS – License Agreement page appears.
14	Select I accept the terms in the license agreement option. Click Next . The Experion PKS – Customer Information and Software Licenses page appears.
15	Enter the appropriate information. Select Yes for Are there licensed TPS components to install? option. Click Next . The Experion PKS – Optional Components page appears.
16	Select the required optional component. Click Next . Note: The dialog box differs based on the optional component you select. The Experion PKS – Server Name page appears.
17	Enter the server name to which the ES-T will communicate. If the server is redundant, select the Server is redundant option. Click Next . The Experion PKS – Confirm Installation Settings page appears.
18	Click Next . The Experion PKS – Product Selection page appears.
19	Click Continue .

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Step	Action
12	From the Summary tab, click Save Configuration and click OK . Note: If the tool determines the key information is not valid, it prompts you to make corrections before the save operation is complete.
13	Click Exit and then Yes to return to the Node Definition Tool , and go to the "Saving the node definition" procedure.

Configuring the station connections for a Flex Station

Use this procedure to define the Station settings for the station connection property files (.stn files).

Step	Action
1	From the Connect dialog box: • Click New Connection, or • Select your type of connection and click Edit Connection.
2	Select one of the following connection types: Static Station: Permanent, dedicated connection type recommended for Flex Stations used by the operators. Rotary Station: Required connection type recommended for users who do not need full-time access to the server, or who need remote access, typically through a modem. Console Station: Standard connection. Complete the connection information for the node on each tab of the Connection Properties dialog box.
3	After entering information in all the tabs, save the connection configuration.
4	Click Exit to return to the Node Definition Tool .
5	If you are creating a new node, the Optional Software dialog box automatically launches. Click Exit and return to Node Definition Tool .

Saving the node definition

Use this procedure to save the node definition. If two users save a node definition to the EMDB at the same time, only one node definition is saved.

Step	Action
1	From the Node Definition Tool, click Save .
2	From the Save dialog box select the database for the node definition file: • Select Save to EMDB (online only) and click OK, or • Select Save to Offline Folder (online or offline) and click OK.
3	If all the information you have entered is valid, Save operation complete appears in the status box. If the information is not valid, a Validation Errors dialog box appears and the Configure Experion Node Installation (EPC Tool) task is highlighted in red.

Creating an options disk

Use this procedure to create a node definition options disk.

Step	Action
1	If you are creating an options disk insert a floppy disk into the drive.
2	From the Node Definition Tool , click Create Install/Commission Options Media .
3	Navigate to the floppy disk drive or other location and click OK .
4	Click Exit when the Status says Save configuration files operation complete .

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Running the commissioning tool

Use this procedure to launch the commissioning task on a restored node.

Step	Action
1	Restart the computer and log on using an account with at least local administrator privileges.
2	Insert the Experion Application media into the disk drive.
3	Insert the node definition options floppy disk into drive A.
4	Select Start > Run and type commissioningtask . Click OK . The Commissioning Task window appears.
5	When prompted, remove the floppy disk from drive A and then click OK to continue. A confirmation window appears.
6	Click Yes to continue. The computer restarts automatically.



TIP

As a final function, the commissioning tool launches the ANCIM Reboot Manager and performs the following:

- Changes the computer name
- Adds or updates the IP address(es)
- Adds the computer to a domain if applicable.

In addition, the ANCIM Reboot Manager performs the following:

- Manages the reboots required after performing the above actions
 - Runs the link domain groups
 - Runs the server rename tools
-

Running Name Change Utility Software

Run the Name Change Utility to change the cluster server name. Use the following procedure to change the server name.

Step	Action
1	Browse to C:\Program Files\Honeywell\Experion PKS\Utilities\namechg
2	Double-click NameChg to launch Name Change Utility.
3	In the Remote Server Configuration Change dialog box, enter the Server Name of the cluster to which the node is connected. Click Perform Change . The following message appears: Application has finished completing the changes!

Configuring FTE – Speed and Duplex settings

Use the following procedure to configure the FTE – Speed and Duplex settings.

Step	Action
1	Click Start > Settings > Control Panel > Network Connections .
2	Click the Primary Supervisory Network Connection .
3	Select Properties > General > Configure . A dialog box appears saying changes have been made to the properties of this connection. It will be lost if you proceed.
4	Click Yes . The Intel(R) PRO/1000 MT Dual Port Network Connection... dialog box appears.
5	Click the Link Speed tab. Select the desired Speed and Duplex. Note: The settings depend on the make/model of the Network Interface Card.

8. Multiple Station load based on common image

8.4 Post deployment steps for Experion nodes

Updating the host file and configuring the LCN address

You must update the host file on both the EPKS server and the Experion node. Perform the following steps on the destination node to update the host file.

Step	Action
1	Browse to c:\WINDOWS\system32\drivers\etc.
2	Open the hosts file in notepad and edit the IP address and hostname of the machine to which the connection must be established.
3	Configure the LCNP (Board0) Settings.
4	Restart both the server and the client machines for the above mentioned steps to take effect.

8.5 Post deployment steps for TPS nodes

Checklist for deploying images on the destination nodes

Perform the following procedures on the destination node after deploying the image.

Step	Task
1	Configure TCP/IP Settings
	REFERENCE - INTERNAL For information about configuring TCP/ICP settings, see "TPS NODE PROCEDURES > Configuring TCP/IP settings" section of the <i>TPS System Implementation Guide for Windows 2003/XP</i> .
2	Configure FTE Service Settings
	REFERENCE - INTERNAL For information about configuring FTE service settings, see "Configuring FTE Service properties > Configure FTE service settings" section of the <i>Fault Tolerant Ethernet Installation and Service Guide</i> .
3	Configure LCNP (Board0) Settings
	REFERENCE - INTERNAL For information about configuring LCNP (Board0) settings, see "TPS NODE PROCEDURES > Configuring LCNP (Board0) settings" section of the <i>TPS System Implementation Guide for Windows 2003/XP</i> .
4	Join the Windows 2003domain if not joined already. This step is required only for TPS GUS, Network GUS, and APP nodes.
	TIP Right-click My Computer and select Properties . Select the Computer Name tab and click Change .
5	Link the domain Windows groups to local Windows groups
	REFERENCE - INTERNAL For information about configuring LCNP (Board0) settings, see "TPS NODE PROCEDURES > Performing LinkDomainGroups" section of the <i>TPS System Implementation Guide for Windows 2003/XP</i> .

8.6 Cloning procedure for non-EBR imaging software

Perform the following steps to prepare master node for cloning for non EBR imaging software.

Step	Action
1	Setup a Master node. For setting up a Master node, see section "Setting up a Master node."
2	Run Sysprep. For running the Sysprep, see "Procedure to run Sysprep."
3	Install Node Cloning Solution software. For installing the node cloning solution software, see section "Installing Experion Backup and Restore software Package."
4	Prepare Master node for cloning. For preparing the master node for cloning, see section "Preparing the Master node for Cloning."
5	Run your node imaging software (usually from a bootable disk or CD) to create a disk image.
6	Restore the Image to a new computer using the same imaging software. Note: You must not restart the node as a script After_Restore_For_Clone.vbs has to be run after the image restoration is complete and before the node restarts.
7	Run Before_Reboot_For_Clone script For information about running Before_Reboot_For_Clone script, see step 14 in the procedure "Deploying the Master Image."
8	Restart the computer with the restored Image. You can log on after the computer restarts.
9	The After_Restore_For_Clone script is run automatically. You can log on after the computer restarts.

8. Multiple Station load based on common image

8.6 Cloning procedure for non-EBR imaging software

Step	Action
10	Configure user-specific and computer-specific settings. Perform the following procedures: • Post deployment steps for Experion nodes • Post deployment steps for TPS nodes

Procedure to run Sysprep

Perform the following steps to run Sysprep on the master node.

Step	Action
1	Logon as local Administrator.
2	Click Start > Run , type cmd , and then click OK .
3	At the command prompt, change to the root folder of drive C, and then type md Sysprep .
4	Insert the Windows operating system CD into the appropriate CD-ROM or DVD-ROM drive, and then open the Deploy.cab file in the \Support\Tools folder. Note: To ensure that you are using the correct version of the Deploy.cab file for your service pack, use the Deploy.cab file that is distributed with that service pack.
5	Copy the Sysprep.exe file and the Setupcl.exe file to the Sysprep folder.
6	If you use the Sysprep.inf file, copy this file to the Sysprep folder also. For the Sysprep tool to function correctly, the Sysprep.exe, the Setupcl.exe, and the Sysprep.inf files must be in the same folder. Refer "Creating the Sysprep.inf answer file using the Setup Manager tool".

8. Multiple Station load based on common image

8.6 Cloning procedure for non-EBR imaging software

Step	Action
7	Sysprep enumerates (or rediscovers) hardware the first time the cloned computer starts, you have to add driver files for some hardware if the drivers do not come with your OS installation files. If this is the case, create a driver directory under c:\sysprep and in the driver's directory create a separate folder for each device that needs drivers installed. For example, if you need to add drivers for your video card and NIC, you could create: c:\sysprep\drivers\video and c:\sysprep\drivers\nic. In each of those folders (..\video, ..\nic) copy the needed driver files. These files must contain not only the driver files, but an .inf and a .cat file for the device to be installed. Do not copy a binary file (.exe) that your hardware manufacturer created to update your hardware. You must indicate in your sysprep.inf file the location of these files.  ATTENTION If the hard drive controller(s) in your destination node(s) are different than your Master node, you must indicate in the sysprep.inf file how to locate the .inf file for those controller(s).
8	Remove the Windows XP operating system CD.
9	At the command prompt, type cd Sysprep to change to the Sysprep folder.
10	To run the Sysprep tool, type the following command at the command prompt: sysprep.exe /clean  ATTENTION Microsoft recommends that after you run the Sysprep.exe file on a reference computer; isolate the reference computer from the local network where the image is subsequently deployed to avoid any NetBIOS conflicts. This permits the Sysprep tool to complete the Setup program, join the domain that is specified, and use the new computer name that is specified in the Sysprep.inf answer file.

Creating the Sysprep.inf answer file using the Setup Manager tool

To install the Setup Manager tool and to create an answer file, follow these steps:

Step	Action
1	Open My Computer , and then open the Support\Tools folder on the Windows OS CD
2	Double-click the Deploy.cab file to open it.
3	On the Edit menu, click Select All .
4	On the Edit menu, click Copy To Folder .
5	Click Make New Folder . Type the name that you want for the Setup Manager folder, and then press ENTER. For example, type setup manager, and then press ENTER.
6	Click Copy .
7	Open the new folder that you created, and then double-click the Setupmgr.exe file. The Windows Setup Manager Wizard starts.
8	Create an answer file. For creating an answer file, see section "Creating the Sysprep.inf answer file."

Creating the Sysprep.inf answer file

Perform the following steps in the Windows Setup Manager wizard to create an answer file.

Step	Action
1	On the Welcome to Sysprep Manager page, click Next . The New or Existing Answer File page appears.
2	Select Create new , and then click Next . The Type of Setup page appears.
3	Select Sysprep Setup , click Next . The Product screen page appears.

8. Multiple Station load based on common image

8.6 Cloning procedure for non-EBR imaging software

Step	Action
4	Select the type of Operating System of the destination node, and then click Next . The License Agreement page appears.
5	Select the default option, and then click Next . The Name and Organization page appears.
6	Enter the Name and Organization details, and then click Next . The Display Settings page appears.
7	Select the display settings, and then click Next . The Time Zone page appears.
8	Select the desired time zone, and then click Next . The Product Key page appears.
9	Enter the Product Key, and then click Next . The Computer Name page appears.
10	Enter the Computer Name , and then click Next . The Administrator Password page appears.
11	Accept the default option, and then click Next . The Networking Components page appears.
12	Accept the default option, and click Next . The Workgroup or Domain page appears.
13	Select Domain , and then for the machine to be part of the domain select Create a computer account in the domain option. Enter the User name and Password , and then click Next . The Telephony page appears.
14	Click Next . The Regional Settings page appears.

8. Multiple Station load based on common image
8.6 Cloning procedure for non-EBR imaging software

Step	Action
15	Click Next . The Languages page appears.
16	Select Western Europe and US , and then click Next . The Install Printers page appears.
17	Click Next . The Run Once page appears.
18	Click Next . The Additional Commands page appears.
19	Click Next . The Identification String page appears.
20	Click Finish . The Setup Manager page appears.
21	Click OK . The Completing Setup Manager page appears.
22	Click Cancel .

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